SFSC-CDNet: A Novel Change Detection Approach for Collaborative Inference of Structure and Semantics in Multitemporal Remote Sensing Images

Shu Qian, Qian Shen, Junli Li, Xin Zhang, Rong Bu, and Changming Zhu

***Abstract*—Semantic change detection (SCD) aims to simultaneously identify change locations and semantic transfer relationships in multitemporal remote sensing images. However, existing difference modeling predominantly emphasizes spatial similarity, often struggling to maintain stability in cross-temporal difference representation while ensuring high-fidelity boundary delineation. This article introduces an end-to-end unified framework, SFSC-CDNet, which facilitates collaborative inference of structure and semantics through spatial–frequency synergistic difference interaction (SFSDI), edge-guided interactive semantic transmission (EGIST), and the bidirectional semantic consistency-guided multitask decoder (SCCM). It incorporates a cross-temporal semantic consistency loss $\mathcal{L}_{\text{cwsc}}$, explicitly constraining the stability of bidirectional semantic mapping, thereby enhancing the model's robustness and boundary fidelity. Experiments conducted on four SCD datasets (SECOND, LevirSCD, Landsat-SCD, WUSU) demonstrate that this method significantly surpasses the current state-of-the-art (SOTA), achieving SeK improvements of 0.87%, 3.88%, 1.68%, and 6.73%, respectively. Boundary-aware evaluation results indicate that SFSC-CDNet consistently performs well in terms of $F_{boundary}$ and Brier score, reflecting its superior boundary delineation capability and more reliable probability calibration. In addition, under comparable parameter conditions, it achieves a reduction in GFLOPs by 72.51%–74.56% compared to the SOTA method, significantly lowering computational costs and confirming that the method establishes a better accuracy-efficiency balance in both bi-temporal and multitemporal inference scenarios. The associated code is available at https://github.com/Brookyway/SFSC-CDNet**

***Index Terms*—Difference modeling, remote sensing, semantic change detection (SCD), semantic consistency.**

Received 11 May 2026; revised 1 June 2026; accepted 15 June 2026. Date of publication 18 June 2026; date of current version 29 June 2026. This work was supported in part by the National Key Research and Development Program of China under Grant 2023YFE0103800, in part by the National Natural Science Foundation of China under Grant 42571346, and in part by Jiangsu Qinglan Project and the Research and Practice Innovation Plan for Postgraduates of Jiangsu Normal University under Grant 2025XKT1060. *(Corresponding author: Changming Zhu.)*

Shu Qian, Qian Shen, Rong Bu, and Changming Zhu are with the School of Geography, Geomatics and Planning, Jiangsu Normal University, Xuzhou 221116, China (e-mail: 2020241644@jsnu.edu.cn; 6020240055@jsnu.edu.cn; 2020241695@jsnu.edu.cn; Zhuchangming@jsnu.edu.cn).

Junli Li is with the State Key Laboratory of Ecological Safety and Sustainable Development in Arid Lands, Xinjiang Institute of Ecology and Geography, CAS, Urumqi 830011, China (e-mail: lijl@ms.xjb.ac.cn).

Xin Zhang is with the Institute of State Key Laboratory of Remote Sensing Science, Institute of Aerospace Information Innovation, CAS, Beijing 100049, China (e-mail: zhangxin000181@aircas.ac.cn).

Digital Object Identifier 10.1109/TGRS.2026.3705004

I. Introduction

THE advancement of Earth observation capabilities has endowed high-resolution remote sensing imagery with broader spatiotemporal coverage and a higher frequency of acquisition. This facilitates a transition in change detection from traditional bi-temporal comparisons to refined analyses under multitemporal continuous observation [1], [2], thereby laying a robust data foundation for delineating land cover/land use (LCLU) change processes [3]. Within this framework, semantic change detection (SCD) utilizes multitemporal remote sensing imagery as input [4], not only pinpointing regions of change but also illustrating the transition of LCLU categories before and after the change. This imbues the change results with distinct semantic connotations. In contrast to binary change detection, which merely indicates changed or unchanged status, SCD elucidates the transfer dynamics and evolutionary trajectories of LCLU [5]. It extends the discrete comparison between two periods to encompass the representation of phased transitions and incremental evolution across multitemporal sequences [6], [7]. This approach offers more structured explanations, attributions, and assessment criteria for refined decision-making in territorial spatial management, urban expansion evaluation, and ecological environment monitoring [8], [9], [10], [11].

In multitemporal high-resolution scenarios, SCD predominantly faces three significant challenges: 1) unstable attribution of discrepancies [12], where variations in imaging conditions and minor spatiotemporal misalignments introduce a substantial number of nonchange discrepancies, thus interfering with the extraction of genuine change cues; 2) degradation of structural representation [13], where fine-grained and complex morphological targets are susceptible to weakened boundaries and local structures during the processes of fusion and resolution restoration; and 3) conflicting objectives in joint learning [14], in which the preference for smoothness and consistency in semantic segmentation and the sensitivity to differences in change discrimination mutually constrain each other within a shared representation framework. Current methods typically pursue three avenues for enhancement: strengthening cross-temporal interaction and contextual modeling, integrating structural information to refine detail restoration, and jointly optimizing localization and class transfer through multitask

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:06:20 UTC from IEEE Xplore. Restrictions apply.

learning. Despite the performance improvements driven by these approaches, several methodological bottlenecks remain. On one hand, the inevitable imaging discrepancies in multitemporal observations render the correspondence between observed discrepancies and surface changes nonunique, and existing paradigms still lack a more robust mechanism for interpreting discrepancies [15]. On the other hand, the integration of spatial details often relies on local enhancement [16]. Such enhancement is predominantly driven by local variations in texture, contrast, and boundary cues. Spurious responses induced by temporal appearance variations and local disturbances, rather than by genuine structural changes, may also be reinforced together with responses truly related to change. Their subsequent aggregation can impair the consistency of structural details and hinder the preservation of boundary and fine structure representation throughout the process. Furthermore, while joint learning facilitates representation sharing [17], the preference disparities between task objectives may still result in suboptimal optimization and objective imbalance.

The aforementioned bottleneck arises from the necessity of concurrently considering differential interpretability, structural preservation, and collaborative inference within an acceptable cost framework in multitemporal SCD. However, existing methods typically optimize only one aspect, sacrificing others in the process. Moreover, as the number of temporal phases increases, the computational and storage demands grow rapidly, complicating the implementation of a holistic multiphase solution. Therefore, a unified paradigm is still required to balance efficiency and accuracy while integrating cross-temporal representation learning, fine-grained structural modeling, and multitask inference. To address these challenges, this article proposes a unified framework for multitemporal SCD that effectively balances fine-grained structural delineation, reliable class-transfer inference, and computational cost.

The main contributions of this article are as follows.

1) We propose SFSC-CDNet, a unified and computationally efficient framework for bi-temporal SCD and its tri-temporal extension. The framework incorporates the intermediate observation as a supervised semantic state in a shared representation space, enabling target first-to-last semantic change inference without multiple independent pairwise networks. Compared with recent state-of-the-art (SOTA) methods, SFSC-CDNet reduces GFLOPs by 72.51%–74.56%.
2) We design the spatial–frequency synergistic difference interaction (SFSDI), which constructs cross-temporal difference features and refines them through parallel spatial- and frequency-domain modeling. By capturing band-wise energy redistribution and adaptively recalibrating discrepancy channels with prompt-guided spectral refinement, SFSDI strengthens semantic change responses and reduces pseudo-change interference under temporal appearance variations.
3) We propose the edge-guided interactive semantic transmission module (EGIST), which guides the selective transmission of cross-temporal information using boundary-related structural evidence. By incorporating Gaussian-smoothed discrepancy cues and edge-related transition responses into spatial gating and cross-gated fusion, EGIST enhances responses to reliable change boundaries while suppressing the diffusion of spurious edge responses caused by texture variations and minor misalignments.
4) We construct a bidirectional semantic consistency-guided multitask decoder (SCCM) and devise a loss function $\mathcal{L}_{\text{cwsc}}$ to explicitly coordinate task pReferences from both decoding and optimization perspectives, thereby alleviating multitask interference.

II. Related Work

A. Cross-Temporal Difference Modeling and Feature Interaction

Intertemporal difference modeling is fundamental to SCD [18], aiming to extract stable and discriminative change evidence from multitemporal images under illumination variations, seasonal differences, and minor registration errors [19]. Daudt et al. [20] introduced fully convolutional Siamese frameworks for change detection (FC-Siam series), which utilize shared-weight encoding and feature differencing to form difference representations. However, these models are sensitive to local texture and radiometric variations. Subsequent research has enhanced cross-temporal interaction from various perspectives, including multiscale fusion and deep supervised discrepancy learning (IFN/DSIFN) [21], as well as the joint spatiotemporal modeling framework SCanNet [22], which employs spatiotemporal constraints to guide semantic change learning. More recently, ChangeFormer [23] leveraged Transformer-based global attention to enhance contextual consistency in high-resolution scenes, while ChangeMamba [24] utilized state-space modeling to capture long-range dependencies with near-linear complexity, achieving strong performance in SCD. Nevertheless, most existing difference modeling methods remain dominated by spatial-domain similarity, where local texture, contrast, and boundary responses serve as the main indicators of changes. These spatial cues are susceptible to weak-texture regions, complex-texture backgrounds, illumination variations, and minor misregistration, resulting in missed detections or false alarms. Frequency-domain cues characterize the energy redistribution of cross-temporal differences and provide additional evidence for structural changes. However, in SCD, frequency modeling should not be limited to global spectral statistics or generic feature enhancement; instead, it should act directly on temporal discrepancy signals to capture frequency-band responses that are truly related to semantic transitions. Conventional frequency-domain designs are commonly applied to original images or feature representations for resolution adaptation or generic feature enhancement, whereas SCD-oriented discrepancy modeling requires the explicit analysis of cross-temporal difference energy. Since true semantic changes and pseudo-change perturbations may exhibit inconsistent energy distributions across frequency bands, discrepancy modeling for SCD requires not only the extraction of spatial and frequency responses, but also the selective refinement of semantic-change-related discrepancy channels while suppressing perturbation-induced responses.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:06:20 UTC from IEEE Xplore. Restrictions apply.

B. Boundary-Aware Change Detection

The changing boundary corresponds to the geometric discontinuities and category transition zones of ground objects. However, it is susceptible to mixed pixels, texture fluctuations, and minor registration errors, which frequently result in inaccuracies such as blurred boundaries, holes, and adhesion. Current research on boundary perception can be broadly classified into three categories: 1) constrained learning targeting boundary quality, wherein Wu and Wei [25] extract boundaries from annotations and devise edge loss and adaptive boundary constraints in conjunction with the Canny operator to enhance localization precision and connectivity; 2) boundary enhancement facilitated by structural priors, where Lin et al. [26] integrate edge priors into multiscale fusion to improve foreground focus and contour completeness; and 3) collaborative learning centered on boundary consistency, where Zhang et al. [27] mitigate the challenge of delineating irregular boundaries through progressive refinement from boundary to center. In the context of SCD, Long et al. [28] introduced BGSNet, and Tang et al. [29] proposed BT-SCD, both employing boundary-guided collaborative learning to reduce the coupling error between boundary detection and classification. Nevertheless, these methods often regard boundaries as supplementary constraints or reinforcement signals during the decoding stage, resulting in inadequate interaction and coupling across temporal discrepancies. Concurrently, edge responses are vulnerable to contamination by high-frequency (HF) unstable responses induced by texture, slight misalignment, and radiometric variations; if these false signals are directly amplified and propagated, they may be exaggerated during fusion, thereby affecting the sensitivity to reliable change boundaries.

C. Multitask Learning for SCD

SCD commonly formulates change localization and bi-temporal semantic assignment as a joint learning problem to alleviate the challenges of supervision scarcity and the optimization difficulties associated with semantic change labels. Prior work either decomposes this task into bi-temporal semantic segmentation and binary change detection, utilizing semantic-change inductive biases for collaborative optimization [30], or explicitly models semantic-change dependencies through contrastive learning or hierarchical structures to enhance "from-to" recognition in complex scenes [31]. However, inherent multitask conflicts persist: segmentation promotes regional smoothness, while change detection emphasizes difference sensitivity [32]. In addition, hard-shared representations may lead to negative transfer [33], resulting in mutual interference and unstable boundary transfer. To address these issues, some methods prioritize conflict suppression to stabilize training, while others model changes through cross-temporal semantic interaction or semantic transfer with structured inference [34]. Overall, most existing approaches continue to rely on hard sharing and auxiliary strategies, which lack an interpretable bidirectional closed-loop coupling at the decoding stage [35]. This limitation restricts stable coordination between change-driven semantic refinement and the suppression of false changes.

III. Methodology

A. SFSC-CDNet Architecture

This article presents a unified end-to-end framework specifically designed for bi-temporal SCD and its tri-temporal extension, referred to as SFSC-CDNet, as illustrated in Fig. 1. The framework consists of five essential components: 1) a backbone encoder; 2) a SFSDI; 3) an edge-guided interactive semantic transmission module (EGIST); 4) a bidirectional SCCM; and 5) a joint loss function.

In the bi-temporal setting, SFSC-CDNet follows a shared feature extraction and target-pair inference paradigm. When extended to the tri-temporal setting, the input observations are expanded from two time phases to three time phases, denoted as I_1, I_2, and I_3. In this work, the multitemporal setting is mainly evaluated using the WUSU tri-temporal dataset. All temporal observations are processed through the weight-shared feature extraction pipeline, and the semantic branch provides supervised semantic predictions for each temporal observation. It should be noted that the tri-temporal extension does not explicitly perform pairwise change prediction for all temporal combinations. Instead, the intermediate observation I_2 is introduced as a supervised semantic state. Through the semantic supervision of I_2, the shared feature extractor is encouraged to learn a more temporally consistent semantic representation across different observations, thereby providing additional intermediate-state information for the final $I_1 \rightarrow I_3$ semantic change prediction. In this way, I_2 contributes to the final prediction through supervised semantic representation learning rather than through an additional independent change-detection branch. Compared with the bi-temporal version, the tri-temporal adaptation mainly extends the input observations and semantic supervision from two time phases to three time phases, while reusing the core difference-interaction and decoding modules for the target first-to-last pair. Since no temporal-specific encoders or multiple independent pairwise branches are introduced, the tri-temporal extension avoids the additional computational burden of constructing separate pairwise change detection networks while preserving the unified target-pair inference paradigm.

In this study, GOST-Mamba is employed as the default encoding backbone, striking a balance between modeling long-range dependencies and ensuring computational efficiency. However, the framework is not limited to a specific paradigm and can be easily replaced with CNN or Transformer encoders, demonstrating its strong transferability and versatility. During the encoding phase, shared-weight feature extraction is performed on images from different temporal phases to construct a multiscale feature pyramid. In the cross-temporal interaction stage, SFSDI explicitly models discrepancies based on paired multiscale features, facilitating structured interaction. EGIST conducts gated filtering and enhancement on boundary-sensitive high-level features, separating structural change evidence from texture perturbations and amplifying spatially continuous boundary responses, which enables change cues to propagate and aggregate across multiscale semantic hierarchies. In the decoding stage, the SCCM decoder is incorporated to establish bidirectional constraints through

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:06:20 UTC from IEEE Xplore. Restrictions apply.

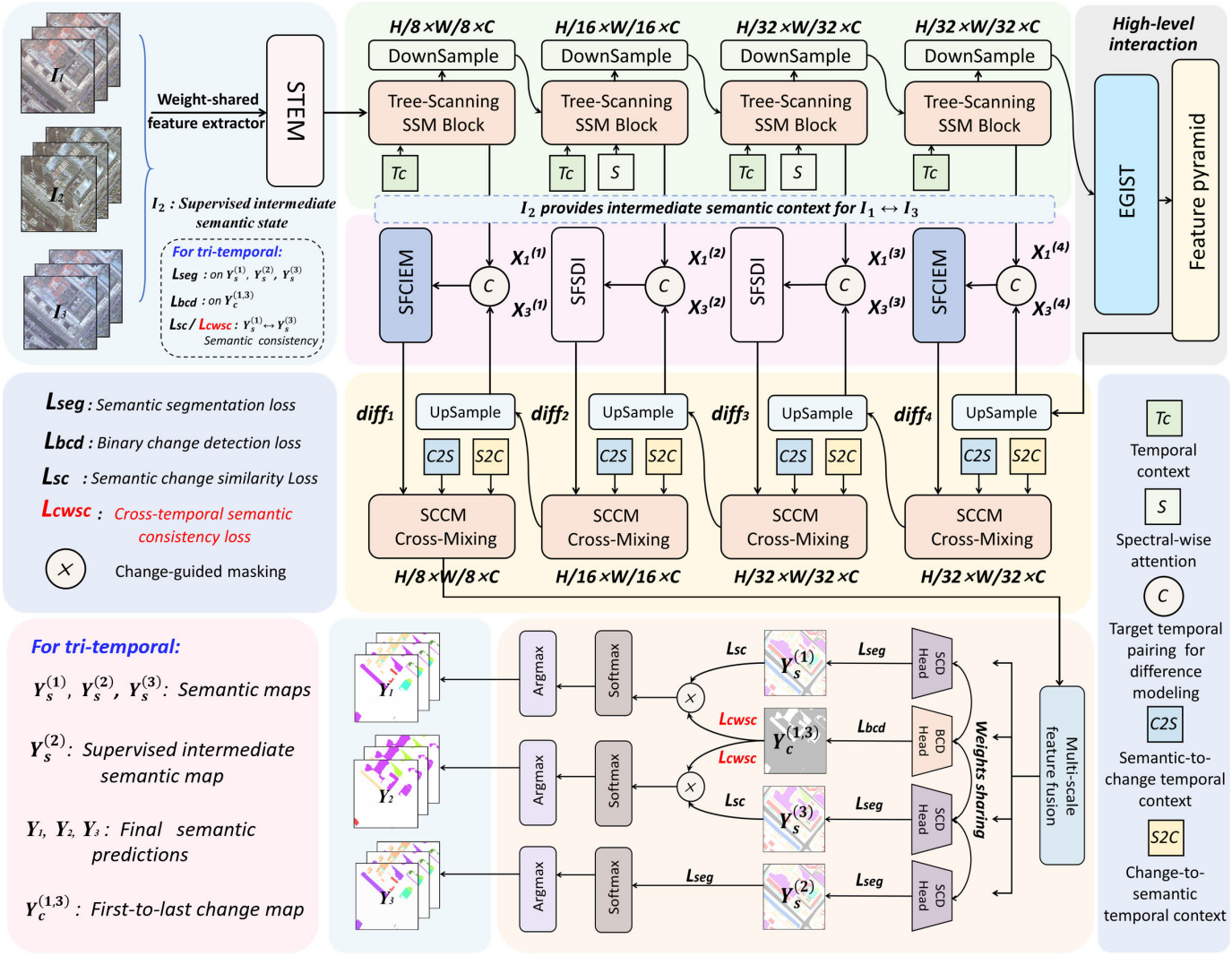


Fig. 1. Overall framework of SFSC-CDNet. The framework is built on a bi-temporal SCD pipeline and is extended to the tri-temporal setting by introducing the intermediate observation I_2. All temporal observations are processed by a weight-shared feature extractor, while the proposed SFSDI, EGIST, and SCCM modules operate on the target first-to-last pair (I_1, I_3) for difference interaction, structure-guided feature transmission, and semantic-change collaborative decoding. In the tri-temporal case, I_2 provides supervised intermediate semantic context, and the final semantic change prediction remains defined on the first-to-last pair $I_1 \rightarrow I_3$. The WUSU tri-temporal setting is the multitemporal instantiation evaluated in this work.

change-guided segmentation semantic refinement (C2S) and semantic consistency suppression of pseudo-changes (S2C), promoting mutual enhancement and suppression between the change branch and the semantic branch. Ultimately, a weighted joint optimization strategy is adopted, integrating semantic segmentation loss, binary change detection loss, and cross-temporal semantic consistency constraints to facilitate collaborative learning across multiple tasks.

B. Spatial–Frequency Synergistic Difference Interaction

Extracting robust change representations with clear boundaries from bi-temporal features is pivotal for enhancing the performance of SCD. However, most existing methods primarily focus on difference modeling within the spatial domain (e.g., convolution, attention, gating, etc.), implicitly relying on local texture statistics to differentiate between genuine and spurious changes [36]. This approach often overlooks the structural priors embedded in the frequency domain and spectrum. Typically, genuine changes are associated with energy redistribution across specific frequency bands, while various LCLU features and types of changes exhibit distinct separability in their spectral distributions. Consequently, we introduce the SFSDI module (Fig. 2), which explicitly integrates frequency-domain response modeling and spectral attention mechanisms into the constructed cross-temporal difference features. The frequency-domain branch operates on discrepancy representations to characterize band-wise energy redistribution induced by temporal discrepancies, with the aim of strengthening semantic-change-related responses and reducing pseudo-change interference caused by temporal appearance variations. Meanwhile, the prompt-guided spectral refinement recalibrates discrepancy channels to further enhance the selectivity of semantic change responses.

Specifically, given the bi-temporal features $F_1^{(j)}, F_2^{(j)} \in \mathbb{R}^{C\times H\times W}$ at the jth layer of the encoder, we first employ a temporal difference operator to explicitly construct the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:06:20 UTC from IEEE Xplore. Restrictions apply.

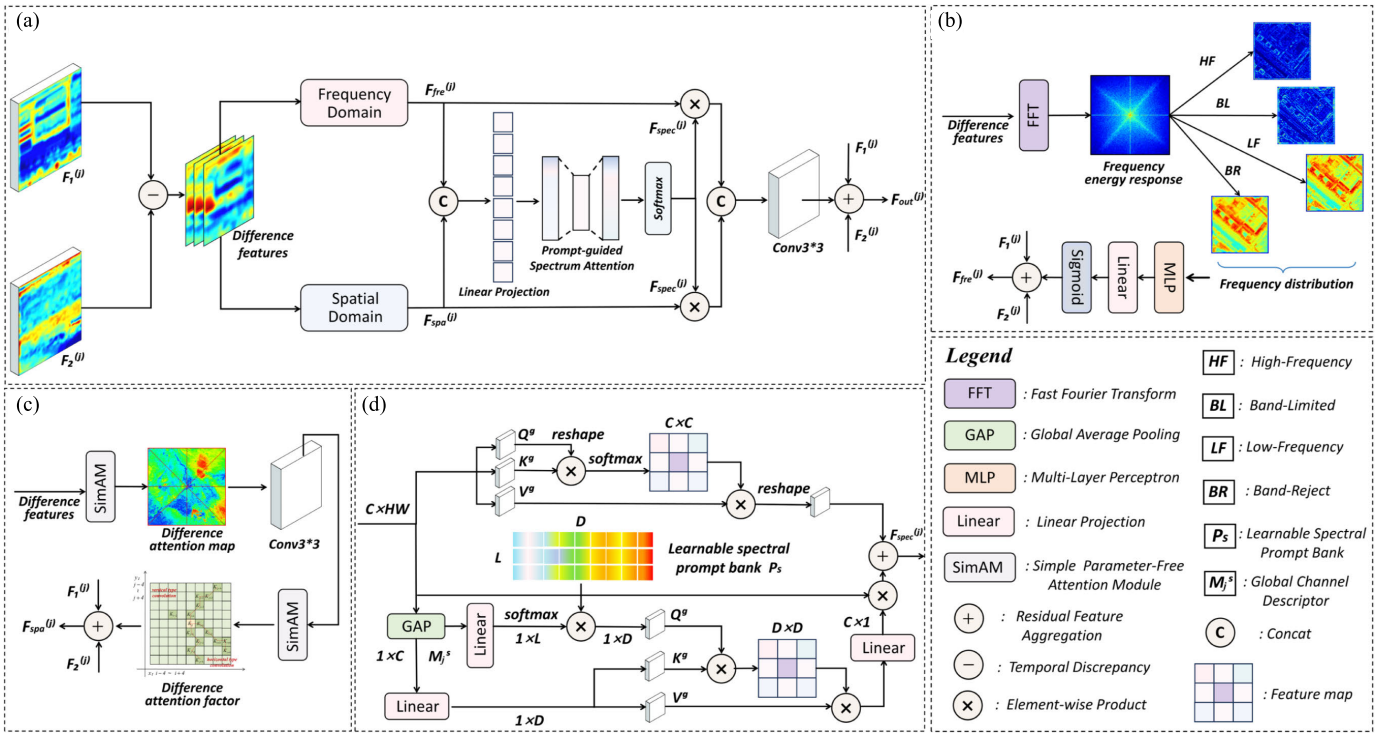


Fig. 2. Overview of the proposed SFSDI module. (a) Overall framework of SFSDI. The bi-temporal features are first transformed into difference features and then processed by the frequency domain branch and the spatial domain branch in parallel, followed by prompt-guided spectrum attention and feature aggregation. (b) Frequency domain, where FFT is applied to obtain the frequency energy response, and predefined frequency bands are adaptively modulated to model change-related spectral redistribution. (c) Spatial domain, where spatial discrepancy responses are enhanced to retain informative local details. (d) Prompt-guided spectrum attention, which consists of a global feature correlation branch and a prompt-induced spectral prior branch. The former captures global channel correlation, while the latter generates adaptive spectral priors from the learnable spectral prompt bank P_s for discrepancy refinement, thereby improving the selectivity of change-related feature responses and reducing redundant interference across heterogeneous representations.

difference feature

$$\Delta^{(j)} = F_1^{(j)} - F_2^{(j)}. \tag{1}$$

As shown in Fig. 2(a), the obtained difference feature is then concurrently fed into the frequency domain branch and the spatial domain branch for complementary structural enhancement. The resulting representations are further refined by the prompt-guided spectrum attention module.

In the frequency domain branch, a fast Fourier transform is applied to the 2-D difference map $\Delta_c^{(j)}(x, y)$ for each channel c to derive complex spectral coefficients $X_c^{(j)}(u, v)$ and the corresponding energy response $E_c^{(j)}(u, v)$

$$\begin{cases} X_c^{(j)}(u,v) = \sum_{x=0}^{H-1}\sum_{y=0}^{W-1} \Delta_c^{(j)}(x,y) \exp\left(-j2\pi\left(\frac{ux}{H}+\frac{vy}{W}\right)\right) \\ E_c^{(j)}(u,v) = |X_c^{(j)}(u,v)|^2. \end{cases} \tag{2}$$

Subsequently, the energy response is aggregated over the predefined frequency band masks $\{M_k(u, v)\}_{k=1}^K$, resulting in the formation of a compact frequency band description vector

$$\begin{aligned} g_{c,k}^{(j)} &= \frac{1}{HW}\sum_{u=0}^{H-1}\sum_{v=0}^{W-1} M_k(u,v)\, E_c^{(j)}(u,v) \\ \mathrm{g}_c^{(j)} &= \left[g_{c,1}^{(j)}, \ldots, g_{c,K}^{(j)}\right]. \end{aligned} \tag{3}$$

Subsequently, $\mathrm{g}_c^{(j)}$ is fed into an MLP to obtain the band selection coefficients, and the spectrum is modulated in a band-aware manner accordingly

$$\begin{cases} \alpha_c^{(j)} = \sigma\left(\phi\left(\mathrm{g}_c^{(j)}\right)\right) \\ \tilde{X}_c^{(j)}(u,v) = \sum_{k=1}^{K} \alpha_{c,k}^{(j)} M_k(u,v)\, X_c^{(j)}(u,v) \end{cases} \tag{4}$$

where $\phi(\cdot)$ denotes the MLP, and $\sigma(\cdot)$ denotes the Sigmoid activation. Here, the MLP is used for nonlinear modeling of the frequency band statistics, whereas the Linear operation in Fig. 2(b) denotes a single linear projection for feature transformation. Ultimately, the frequency-domain enhanced features are acquired by transforming back to the spatial domain via inverse transformation

$$F_{fre}^{(j)}(x,y) = \frac{1}{HW}\sum_{u=0}^{H-1}\sum_{v=0}^{W-1} \tilde{X}_c^{(j)}(u,v) \exp\left(j2\pi\left(\frac{ux}{H}+\frac{vy}{W}\right)\right). \tag{5}$$

In the spatial branch shown in Fig. 2(c), significance modulation is introduced on $\Delta^{(j)}$ to enhance informative discrepancy responses. Specifically, the channel mean and variance are first calculated, and the pixel-level response, attention weight, and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:06:20 UTC from IEEE Xplore. Restrictions apply.

spatially enhanced feature are obtained as

$$\begin{cases} e_c(x,y) = \dfrac{\left(\Delta_c^{(j)}(x,y) - \mu_c\right)^2}{4\left(\sigma_c^2 + \varepsilon\right)} + \dfrac{1}{2} \\ A_c(x,y) = \sigma\left(e_c(x,y)\right) \\ F_{spa}^{(j)} = A_c \otimes \Delta^{(j)} \end{cases} \tag{6}$$

where μ_c and σ_c^2 denote the mean and variance of the cth channel, respectively, and $\otimes$ denotes element-wise product. In this way, the spatial branch selectively enhances informative spatial discrepancy responses and preserves effective local details.

Meanwhile, prompt-guided spectrum attention is conducted in the channel dimension, as shown in Fig. 2(d). Specifically, $\Delta^{(j)}$ is flattened into $\mathrm{Z} \in \mathbb{R}^{C\times HW}$, and the module consists of two cooperative branches. In the global feature correlation branch, linear projections are applied to derive Q^g, K^g, and V^g from which the global channel correlation response is computed

$$\begin{cases} \mathrm{S} = \mathrm{Softmax}\left(\dfrac{\mathrm{Q}^g\left(\mathrm{K}^g\right)^\top}{\sqrt{d}}\right) \\ \mathrm{Z}_g = \mathrm{S}\mathrm{V}^g. \end{cases} \tag{7}$$

In the prompt-induced spectral prior branch, a global channel descriptor M_j^s is extracted by global average pooling and used to generate the prompt weights for the learnable spectral prompt bank P_s. The resulting prompt representation provides adaptive spectral priors for the subsequent refinement

$$\begin{cases} \mathrm{r}_s = \mathrm{Softmax}\left(\mathrm{W}_p M_j^s\right) P_s \\ \mathrm{Z}_p = \mathrm{Linear}\left(\mathcal{A}_p\left(\mathrm{r}_s\right)\right) \\ F_{spec}^{(j)} = \mathrm{reshape}\left(\mathrm{Z}_g + \mathrm{Z}_p\right) \end{cases} \tag{8}$$

where r_s denotes the prompt-induced spectral prior generated from M_j^s and P_s, and $\mathcal{A}_p(\cdot)$ denotes the prompt-guided attention mapping, and the subsequent Linear operation maps the prompt-induced response for integration with the global correlation response. In this module, the global correlation branch captures the input-dependent channel relationships from the difference representation, while the prompt-induced spectral prior branch provides adaptive spectral guidance derived from the learnable prompt bank. Their collaboration enables the correlation response to be calibrated by spectral priors, thereby reducing redundant coupling across heterogeneous representations and improving the selectivity of change-related feature responses.

Finally, the outputs of the frequency domain branch, the spatial domain branch, and the prompt-guided spectrum attention module are integrated to form the refined difference representation

$$F_{out}^{(j)} = \mathcal{F}\left(F_{fre}^{(j)}, F_{spa}^{(j)}, F_{spec}^{(j)}\right) \tag{9}$$

where $\mathcal{F}(\cdot)$ denotes the feature aggregation operation.

C. Edge-Guided Interactive Semantic Transmission

In the context of SCD, prediction errors are frequently concentrated around changing boundaries and fine-structured regions. Such regions contain object contours, semantic transition zones, and local geometric details, but are also vulnerable to radiometric discrepancies, texture fluctuations, mixed pixels, and minor misalignment. When cross-temporal fusion is dominated by amplitude differences or local texture cues, genuine boundary responses may be weakened, whereas pseudo-change responses induced by irrelevant appearance variations may be propagated across the two temporal feature streams [37]. To mitigate this problem, EGIST introduces an edge-guided gated interaction mechanism for cross-temporal feature transmission, as shown in Fig. 3. Specifically, a difference cue is first constructed from the aligned bi-temporal features and is then used to generate two complementary guidance signals: a spatial gate for location-aware modulation and an enhanced difference representation for boundary-aware guidance. In the edge enhancement branch, Gaussian smoothing is employed to suppress isolated HF disturbances caused by local texture variations, while the edge response emphasizes abrupt variations along object contours and semantic transition regions. These guidance signals are further incorporated into cross-gated fusion, enabling reliable boundary-related information to be selectively exchanged between the temporal feature streams while suppressing the propagation of noisy pseudo-change responses.

Initially, following channel alignment, the bi-temporal features $X, Y \in \mathbb{R}^{B\times C\times H\times W}$ are concatenated and projected by BottConv to obtain two compact temporal representations, denoted as x_p and y_p. Based on these aligned features, a difference cue d_y is constructed and then fed into the spatial gate and edge enhancement branches

$$\begin{cases} \left[x_p, y_p\right] = \mathrm{Split}\left(\mathrm{BottConv}\left(\mathrm{Cat}\left(X, Y\right)\right)\right) \\ d_y = \mathcal{M}\left(x_p, y_p\right) \end{cases} \tag{10}$$

where $x_p, y_p, d_y \in \mathbb{R}^{B\times C_m\times H\times W}$, and $\mathcal{M}(\cdot)$ denotes the discrepancy operator corresponding to the discrepancy mode M0/M1/M2/M3 in Fig. 3(a).

As shown in Fig. 3(b), the spatial gate branch provides location-aware modulation for cross-temporal interaction by estimating a gate map $g \in \mathbb{R}^{B\times 1\times H\times W}$ from the discrepancy cue d_y. The gate is then applied to the two temporal representations in the residual scaling form $1+g$, yielding the spatially gated features x_{out} and y_{out}

$$\begin{cases} g = \mathrm{Dropout2D}\left(\mathrm{Smooth}\left(\sigma\left(\mathrm{SpatialGate}\left(d_y\right)\right)\right)\right) \\ x_{out} = (1+g)\odot x_p \\ y_{out} = (1+g)\odot y_p \end{cases} \tag{11}$$

where $x_{out}, y_{out} \in \mathbb{R}^{B\times C_m\times H\times W}$, and $\odot$ denotes element-wise multiplication with channel-wise broadcasting.

The edge enhancement branch further transforms the same discrepancy cue d_y into boundary-aware structural evidence. As illustrated in Fig. 3(c), Gaussian smoothing is first applied to reduce isolated HF disturbances caused by texture fluctuations and minor misalignment. For any batch b, channel c, and spatial position (i, j), the Gaussian kernel K_σ and the smoothed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:06:20 UTC from IEEE Xplore. Restrictions apply.

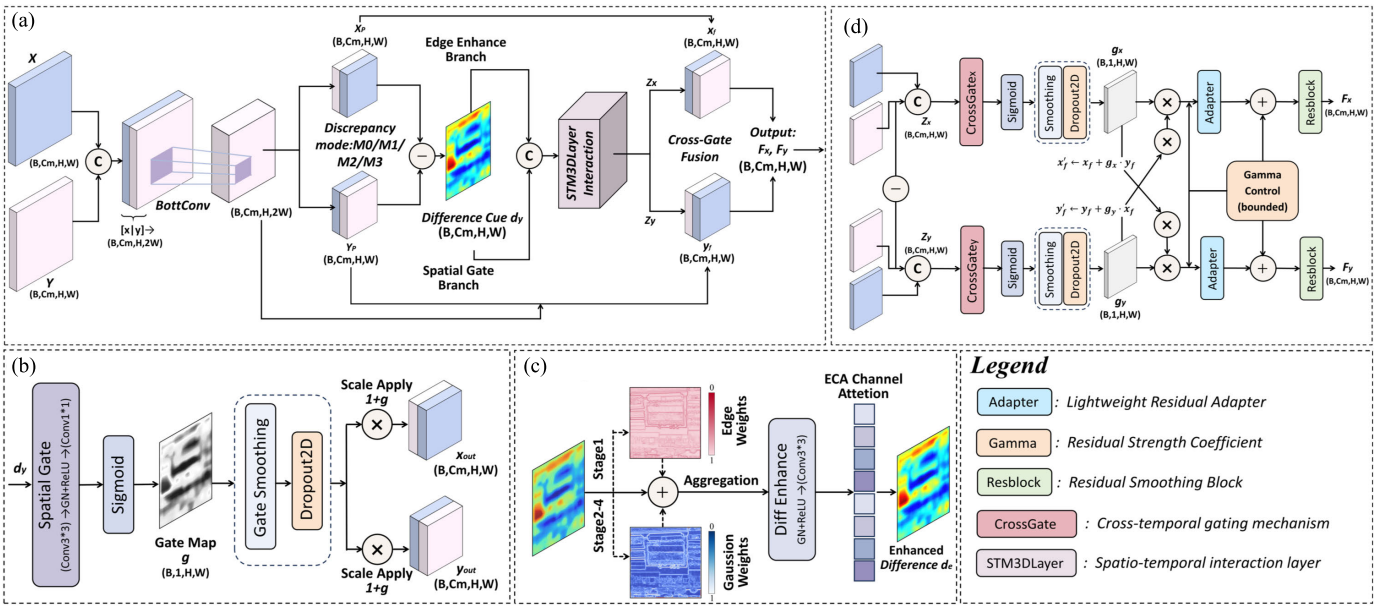


Fig. 3. Overview of the proposed EGIST module. (a) EGIST employs the difference cue d_y derived from the aligned bi-temporal features to guide structure-aware cross-temporal feature interaction. (b) Spatial gate branch generates g to emphasize structurally sensitive regions. (c) Edge enhancement branch aggregates edge and Gaussian responses from different stages to obtain the enhanced difference representation d_e. (d) Resulting interaction representations z_x and z_y are used in cross-gated fusion to regulate bidirectional feature exchange and produce the enhanced outputs F_x and F_y.

discrepancy response are defined as

$$\begin{cases} K_\sigma(m,n) = \dfrac{1}{2\pi\sigma^2}\exp\left(-\dfrac{m^2+n^2}{2\sigma^2}\right) \\ G_{b,c}(i,j) = \displaystyle\sum_{m=-M}^{M}\sum_{n=-N}^{N} d_{y,b,c}(i-m,j-n)\,K_\sigma(m,n). \end{cases} \tag{12}$$

Based on the smoothed response G, a first-order finite difference is used to estimate the edge response, which highlights abrupt variations along object contours and semantic transition regions

$$E_{b,c}(i,j) = \left[\left(\Delta_x G_{b,c}(i,j)\right)^2 + \left(\Delta_y G_{b,c}(i,j)\right)^2\right]^{1/2} \tag{13}$$

where Δ_x and Δ_y denote the finite differences along the horizontal and vertical directions, respectively.

The smoothed response G and the edge response E are then aggregated and refined by the DiffEnhance block and ECA channel attention to obtain the enhanced difference representation d_e

$$d_e = \mathrm{ECA}\left(\mathrm{DiffEnhance}\left(G + \lambda E\right)\right) \tag{14}$$

where λ controls the strength of edge emphasis. The enhanced difference representation $d_e \in \mathbb{R}^{B\times C_m\times H\times W}$ preserves boundary-sensitive structural evidence while reducing the direct amplification of unreliable HF responses.

After obtaining the spatial gate g and the enhanced difference representation d_e, EGIST integrates them with the temporal representations through STM3DLayer to generate two branch-specific interaction representations, denoted as z_x and z_y. As shown in Fig. 3(d), these two branch-specific representations are then used to generate the cross-temporal gates g_x and g_y

$$\begin{cases} g_x = \mathrm{CrossGate}_x(z_x) \\ g_y = \mathrm{CrossGate}_y(z_y) \end{cases} \tag{15}$$

where $g_x, g_y \in \mathbb{R}^{B\times 1\times H\times W}$. The gate g_x controls the information transferred from the y-feature stream to the x-feature stream, whereas g_y controls the reverse transfer. For clarity, the two spatially gated temporal feature streams used for cross-gated feature exchange are denoted as x_f and y_f, where $x_f = x_{out}$ and $y_f = y_{out}$. The gated feature exchange yields the updated features

$$\begin{cases} x'_f = x_f + g_x \cdot y_f \\ y'_f = y_f + g_y \cdot x_f. \end{cases} \tag{16}$$

The transferred information is weighted by the corresponding gate, allowing structurally reliable regions to receive stronger complementary responses from the other temporal feature stream. To further regulate the contribution of the branch-specific interaction representations, a bounded coefficient $\gamma = \tanh(\eta)$ is introduced. The final outputs are obtained by

$$\begin{cases} F_x = \mathrm{Resblock}\left(x'_f + \gamma.\mathrm{Adapter}(z_x)\right) \\ F_y = \mathrm{Resblock}\left(y'_f + \gamma.\mathrm{Adapter}(z_y)\right). \end{cases} \tag{17}$$

D. Semantic Consistency-Guided Multitask Decoder

During the decoding phase, multitask learning can enhance data generalization capability through shared representations. However, in task combinations with notably distinct objectives and supervisory modalities, such as segmentation and change detection, it often leads to pronounced negative transfer [38].

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:06:20 UTC from IEEE Xplore. Restrictions apply.

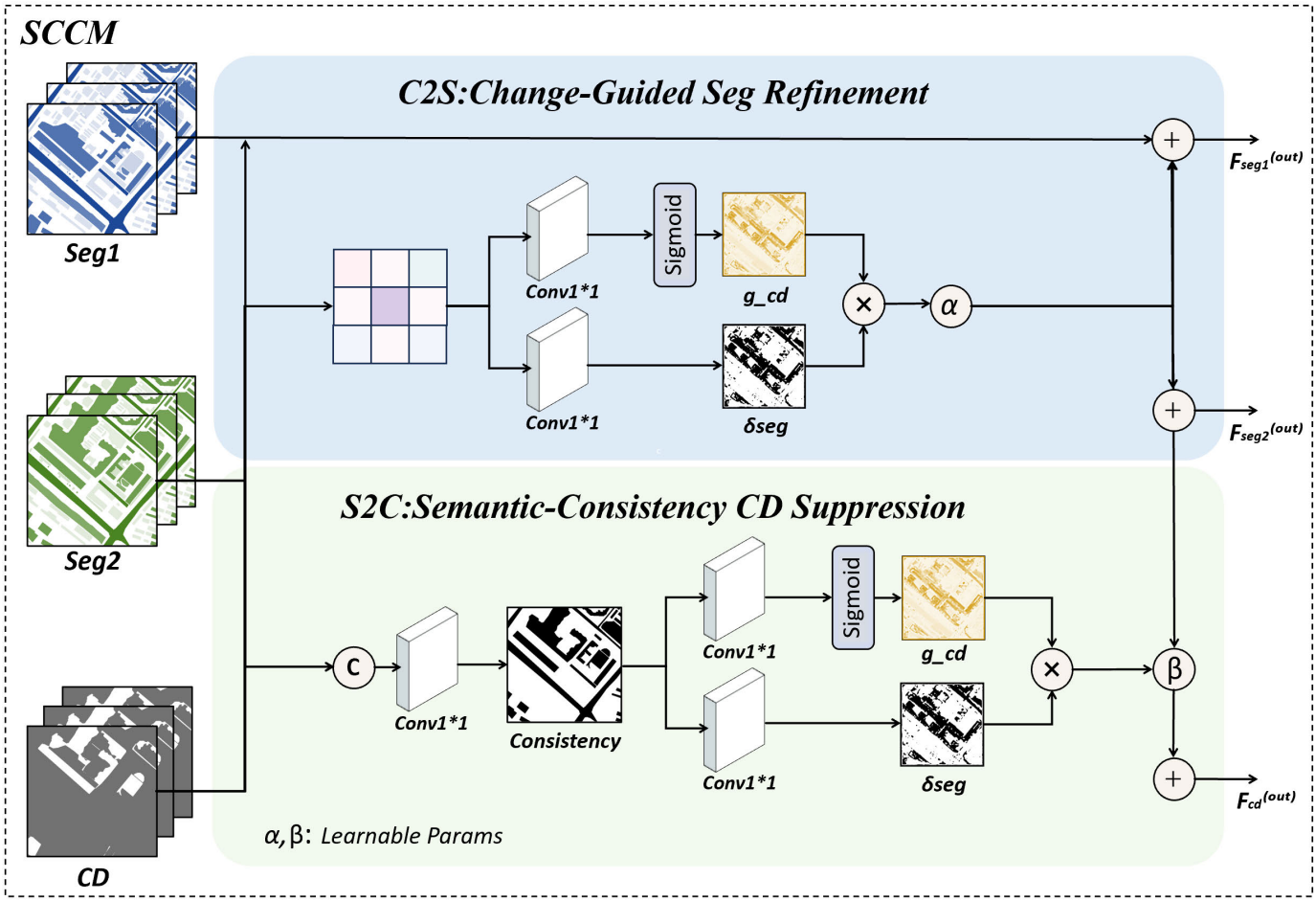


Fig. 4. Structure of the proposed SCCM decoder. SCCM consists of two branches, namely, C2S and S2C. The learnable coefficients α and β regulate the corresponding residual correction paths.

The segmentation task prioritizes regional semantic coherence, favoring the generation of smooth boundaries, while the change detection task emphasizes intertemporal discrepancies and structural discontinuities, focusing more on boundary disruptions and fragmented alterations. The differences between these two tasks in gradient direction and representation preference can easily induce bidirectional interference at the decoding stage, where semantics are disrupted by differential noise and changes are assimilated through semantic smoothing. To address these issues, we introduce the SCCM decoder (Fig. 4), which reconfigures the conventional hard sharing fusion into an interpretable bidirectional constraint: on one hand, it leverages intertemporal semantic consistency to explicitly mitigate semantic drift noise within the change branch; on the other hand, it employs credible change evidence to inversely enhance the boundaries and details of the segmentation branch.

Specifically, suppose the two-phase segmentation features and change features at the decoding end are represented as $F_{s1}, F_{s2}, F_{cd} \in \mathbb{R}^{C\times H\times W}$, respectively. These features undergo two dual pathways, namely, C2S and S2C, and with the aid of learnable coefficients, bidirectional coupling correction with adjustable intensity is implemented. Within the C2S pathway, the change features are initially utilized to generate a change gating mechanism and segmentation residuals, thereby selectively enhancing segmentation details

$$g_{cd}(i,j) = \sigma\left(\sum_{c=1}^{C} w_c^{(g)} F_{cd,c}(i,j)\right)$$
$$\Delta_s(i,j) = \sum_{c=1}^{C} w_c^{(\Delta)} F_{cd,c}(i,j). \tag{18}$$

Then, the gated-modulated residual is integrated into the two-phase segmentation branch, where α serves as a learnable parameter

$$F_{s1}^{out}(i,j) = F_{s1}(i,j) + \alpha g_{cd}(i,j)\Delta_s(i,j)$$
$$F_{s2}^{out}(i,j) = F_{s2}(i,j) + \alpha g_{cd}(i,j)\Delta_s(i,j). \tag{19}$$

In the S2C pathway, a semantic consistency graph is constructed by leveraging two-phase segmentation features, and a point-wise cosine similarity metric is employed for measurement

$$\mathrm{Cons}(i,j) = \frac{\sum_{c=1}^{C} F_{s1,c}(i,j) F_{s2,c}(i,j)}{\sqrt{\sum_{c=1}^{C} F_{s1,c}^2(i,j)}\sqrt{\sum_{c=1}^{C} F_{s2,c}^2(i,j)}}$$
$$g_s(i,j) = \sigma(a\mathrm{Cons}(i,j) + b). \tag{20}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:06:20 UTC from IEEE Xplore. Restrictions apply.

Consequently, we implement a consistency-gated residual correction on the change characteristics (here, β represents a learnable parameter, and $\Delta_{cd}(i,j) = \tilde{w}\ \mathrm{Cons}(i,j)$ serves to provide a consistency-driven noise suppression term)

$$F_{cd}^{out}(i,j) = F_{cd}(i,j) + \beta g_s(i,j)\,\Delta_{cd}(i,j). \tag{21}$$

E. Loss Function

The proposed approach employs multitask joint training to concurrently optimize semantic segmentation and binary change detection across two temporal phases. The overall objective function is formulated as follows:

$$\mathcal{L} = \mathcal{L}_{\mathrm{seg}} + \lambda_1 \mathcal{L}_{\mathrm{bcd}} + \lambda_2 \mathcal{L}_{\mathrm{sc}} + \lambda_3 \mathcal{L}_{\mathrm{cwsc}}. \tag{22}$$

Among them, $\mathcal{L}_{seg}, \mathcal{L}_{bcd}$, and $\mathcal{L}_{sc}$ are employed to supervise the discriminative capabilities of semantic segmentation, binary change detection, and semantic change representation, respectively. Furthermore, in order to establish consistent training constraints that align with the bidirectional coupling correction mechanism of the SCCM decoder, we have devised a cross-temporal semantic consistency loss, denoted as $\mathcal{L}_{cwsc}$, and conducted joint optimization. The SCCM decoder facilitates bidirectional information exchange between the segmentation and change branches. At the optimization level, $\mathcal{L}_{cwsc}$ provides explicit regularization for cross-temporal semantic consistency, ensuring that consistent signals are stably integrated into the joint learning process of multiple tasks. This, in turn, enhances the synergistic effect between segmentation and change detection, while also improving optimization stability during the decoding phase. Specifically, let the semantic probabilities for two temporal phases be represented as $P_1(i,j,k), P_2(i,j,k)$, and the change mask as $M(i,j) \in \{0,1\}$. The cross-temporal semantic difference is defined using symmetric Kullback–Leibler (KL) divergence

$$D(i,j) = \frac{1}{2}\sum_{k=1}^{K}\left[P_1(i,j,k)\log\frac{P_1(i,j,k)}{P_2(i,j,k)} + P_2(i,j,k)\log\frac{P_2(i,j,k)}{P_1(i,j,k)}\right]. \tag{23}$$

The pixel confidence is defined as the average of the maximum class probabilities across the two time phases

$$w(i,j) = \frac{1}{2}\left(\max_k P_1(i,j,k) + \max_k P_2(i,j,k)\right). \tag{24}$$

The weighted term is then defined as

$$\mathcal{L}_{\mathrm{cwsc_weight}} = \frac{1}{|\Omega_0|}\sum_{(i,j)\in\Omega_0}(1 - w(i,j))\,D(i,j). \tag{25}$$

The $\mathcal{L}_{cwsc}$ is defined as follows:

$$\mathcal{L}_{\mathrm{cwsc}} = \underbrace{w_1\mathcal{L}_{\mathrm{cwsc_weight}}}_{\text{Confidence weighting}} + \underbrace{w_2\mathcal{L}_{\mathrm{cwsc_same}}}_{\text{Unchanged consistency}} + \underbrace{w_3\mathcal{L}_{\mathrm{cwsc_diff}}}_{\text{Change de-homogenization}} + \underbrace{w_4\mathcal{L}_{\mathrm{cwsc_dilate}}}_{\text{Boundary robustness}}. \tag{26}$$

Among these, $\mathcal{L}_{\mathrm{cwsc_same}}$ imposes constraints on the cross-temporal semantics within nonchanging regions, $\mathcal{L}_{\mathrm{cwsc_diff}}$ employs de-assimilation constraints on changing regions to retain authentic change discrepancies, $\mathcal{L}_{\mathrm{cwsc_dilate}}$ offers pixel-level tolerance within boundary neighborhoods, and $\mathcal{L}_{\mathrm{cwsc_weight}}$ adaptively reduces weights for low-confidence pixels to bolster training stability. In Section IV-E, we performed a sensitivity analysis on the w value and the weights associated with $\mathcal{L}_{cwsc}$.

IV. Experiments and Results

A. Datasets

1) SECOND: SECOND [39] is a bi-temporal aerial dataset designed for SCD, comprising 4662 geometrically registered image pairs. Each image has dimensions of 512×512 pixels, with a spatial resolution ranging from 0.5 to 3 m per pixel, and encompasses urban areas including Hangzhou, Chengdu, and Shanghai. The annotations adhere to land-cover semantics, categorizing the data into six classes: nonvegetated surfaces, trees, low vegetation, water, buildings, and playgrounds. In addition, a nonchange mask is provided to indicate regions that remain semantically unchanged. In this study, we adhere to the official public protocol for training and testing, thereby ensuring both reproducibility and comparability.

2) LevirSCD: LevirSCD [40] focuses on urban SCD, emphasizing a more granular semantic category framework and a broader representation of change combinations in comparison to SECOND. The dataset is organized into 256×256 patches, with a spatial resolution of approximately 1–2 m per pixel, covering a temporal range from 2010 to 2019. In terms of annotation, it encompasses 16 distinct change categories and incorporates approximately 210 specific change types, making it particularly suitable for evaluating the model's proficiency in discerning nuanced differences among similar categories.

3) Landsat-SCD: Landsat-SCD [41] extends the application of SCD to coarser-resolution and long-term remote sensing scenarios. The dataset comprises 8468 image pairs, each with dimensions of 416×416 pixels at a resolution of 30 m, covering the period from 1990 to 2020. Due to the large pixel size and the prevalence of mixed pixels, changes are typically manifested as macro-level land-cover transformations. Consequently, this dataset is particularly well-suited for evaluating semantic stability at medium to low resolution, as well as for assessing robustness and generalization across cross-temporal domain shifts.

4) WUSU: In the context of Wuhan's urbanization, WUSU [42] offers tri-temporal, precisely co-registered multispectral fused imagery derived from the GF-2 satellite. Following orthorectification, fusion, and cross-temporal registration, the dataset produces four-band images with an approximate resolution of 1 m. Each patch is 512×512 pixels, collectively covering an area of approximately 80 km^2. This dataset encompasses three time phases: 2015, 2016, and 2018, which facilitates the evaluation of continuity in changes across these stages.

B. Experimental Setup

1) Implementation Details: Experiments were conducted on a Linux platform. The models were implemented in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:06:20 UTC from IEEE Xplore. Restrictions apply.

PyTorch and trained end-to-end on a single NVIDIA RTX 4090D GPU utilizing the Adam optimizer ($\beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 1 \times 10^{-8}$). The initial learning rate was 5×10^{-4} with weight decay 1×10^{-4}. We utilized a batch size of 4 for both training and validation processes, conducted training over 100 epochs, and established a change threshold of 0.5. Data loading was performed using four workers with shuffling enabled.

2) Evaluation Metrics: To evaluate the accuracy and robustness of our method for SCD, we report a unified set of metrics on four datasets, which include overall accuracy (OA), mean intersection over union (mIoU), semantic change Kappa (SeK), F_{bcd}, F_{scd}, recall, and precision. OA quantifies overall pixel accuracy, while mIoU assesses segmentation quality amidst class overlap. SeK evaluates the consistency of separating changed semantic classes from the unchanged background. In addition, for change localization, we report mIoU_{lc} and Kappa to provide a comprehensive evaluation from both regional overlap and statistical agreement perspectives. To enhance the capture of boundary and detail fidelity, we also introduce boundary-aware metrics, $F_{boundary}$ and the Brier score. Given the pixel-level confusion matrix $Q = [q_{ij}] \in \mathbb{N}^{C\times C}$, where q_{ij} denotes the number of pixels with ground-truth class j predicted as i, the total number of pixels is $N = \sum_{i=1}^{C}\sum_{j=1}^{C} q_{ij}$. Let class index 0 be no-change, and indices $1, \ldots, C-1$ denote changed semantic classes. The formulas are as follows:

$$\text{OA} = \frac{\sum_{c=0}^{C-1} q_{cc}}{\sum_{i=0}^{C-1}\sum_{j=0}^{C-1} q_{ij}} \tag{27}$$

$$\text{mIoU} = \frac{1}{C}\sum_{c=0}^{C-1} \frac{q_{cc}}{\left(\sum_{j=0}^{C-1} q_{cj}\right) + \left(\sum_{i=0}^{C-1} q_{ic}\right) - q_{cc}} \tag{28}$$

$$\text{Prec.} = \frac{TP}{TP + FP} \tag{29}$$

where TP denotes pixels that have truly undergone change and are predicted as changed, whereas FP denotes pixels that have truly remained unchanged but are predicted as changed

$$\text{Rec.} = \frac{TP}{TP + FN} \tag{30}$$

where FN denotes the pixels that have undergone actual changes but are predicted to remain unchanged

$$F_{bcd} = \frac{2 \cdot \text{Prec.}_{bcd} \cdot \text{Rec.}_{bcd}}{\text{Prec.}_{bcd} + \text{Rec.}_{bcd}} \tag{31}$$

$$F_{scd} = \frac{2 \cdot \text{Prec.}_{scd} \cdot \text{Rec.}_{scd}}{\text{Prec.}_{scd} + \text{Rec.}_{scd}} \tag{32}$$

$$mIoU_{lc} = \frac{1}{2}\left(\frac{q_{00}}{\left(\sum_j q_{0j}\right) + \left(\sum_i q_{i0}\right) - q_{00}} + \frac{\sum_{i=1}^{C-1}\sum_{j=1}^{C-1} q_{ij}}{\left(\sum_{i=1}^{C-1}\sum_j q_{ij}\right) + \left(\sum_{j=1}^{C-1}\sum_i q_{ij}\right) - \sum_{i=1}^{C-1}\sum_{j=1}^{C-1} q_{ij}}\right) \tag{33}$$

$$kappa = \frac{\frac{1}{N}\sum_{c=0}^{C-1} q_{cc} - \sum_{c=0}^{C-1}\left(\frac{\sum_{j=0}^{C-1} q_{cj}}{N}\right)\left(\frac{\sum_{i=0}^{C-1} q_{ic}}{N}\right)}{1 - \sum_{c=0}^{C-1}\left(\frac{\sum_{j=0}^{C-1} q_{cj}}{N}\right)\left(\frac{\sum_{i=0}^{C-1} q_{ic}}{N}\right)}. \tag{34}$$

SeK is introduced to reduce the dominant impact of the large proportion of unchanged pixels on consistency evaluation, and it jointly considers change-region overlap to assess the model's ability to distinguish changed areas and their semantic categories. Let κ_{n0} be the Kappa coefficient recomputed by nullifying the unchanged-class entry q_{00} in the confusion matrix Q. Meanwhile, all changed categories $\{1, \ldots, C-1\}$ are merged into a single changed class, and the corresponding IoU is computed as IoU_{chg}. The definition is given as follows:

$$\text{SeK} = \frac{\kappa_{n0} \exp\left(\text{IoU}_{chg}\right)}{e}. \tag{35}$$

$F_{boundary}$ and the Brier score are statistical metrics for evaluating boundary sensitivity and reliability. Specifically, $F_{boundary}$ measures boundary localization consistency by matching the predicted boundary set B_p with the ground-truth boundary set B_g within a tolerance τ (default 5 pixels), jointly reflecting boundary precision and recall. In contrast, the Brier score computes the mean squared error between the predicted probability p_n and the ground-truth label y_n, defined as follows:

$$\begin{aligned} n_p &= |\, x \in B_p : \exists y \in B_g, d(x,y) \le \tau \,| \\ n_g &= |\, y \in B_g : \exists x \in B_p, d(x,y) \le \tau \,| \\ F_{\text{boundary}} &= \frac{2\left(n_p + n_g\right)}{|B_p| n_g + |B_g| n_p} \end{aligned} \tag{36}$$

$$\text{BrierScore} = \frac{1}{N}\sum_{n=1}^{N} (p_n - y_n)^2. \tag{37}$$

C. Comparison Methods

To validate the efficacy of the proposed method, this study selects 15 representative classic and SOTA change detection networks for comparative analysis, encompassing three frameworks: CNN, Transformer, and Mamba. The CNN methods include BiSRNet [43], EGMSNet [44], UNet-3D [45], FC-Siam-diff [20], FC-Siam-conv [20], HRSCD_str3 [46], HRSCD_str4 [46], SSCD-l [43], and TED [22]. The Transformer methods comprise SCanNet [22], ChangeFormerV5 [23], ChangeFormerV6 [23], S-cCDNet [47], and SwinSUNet [48]. The Mamba methods consist of FEMCD [49] and the SFSC-CDNet proposed in this article.

All comparative methods have publicly available implementations. For a fair comparison, we reproduced each method in a unified experimental environment and retrained and evaluated it on the corresponding datasets, employing consistent data preprocessing, loss functions, augmentation strategies, and hyperparameter settings as utilized in our study.

D. Comparative Experimental Results and Analysis

1) Experiments on Second: On the SECOND dataset, we perform both quantitative and qualitative comparisons of SFSC-CDNet with representative methods (Table I). SFSC-CDNet demonstrates robust performance across core metrics, achieving 73.11% mIoU, 23.58% SeK, 73.98% F_{bcd}, and 62.82% F_{scd}. When compared to SCanNet, the strongest baseline, SFSC-CDNet shows improvements of 0.87% in SeK, 0.69% in F_{scd}, and 0.27% in mIoU, indicating enhanced segmentation quality in changed regions and more consistent

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:06:20 UTC from IEEE Xplore. Restrictions apply.

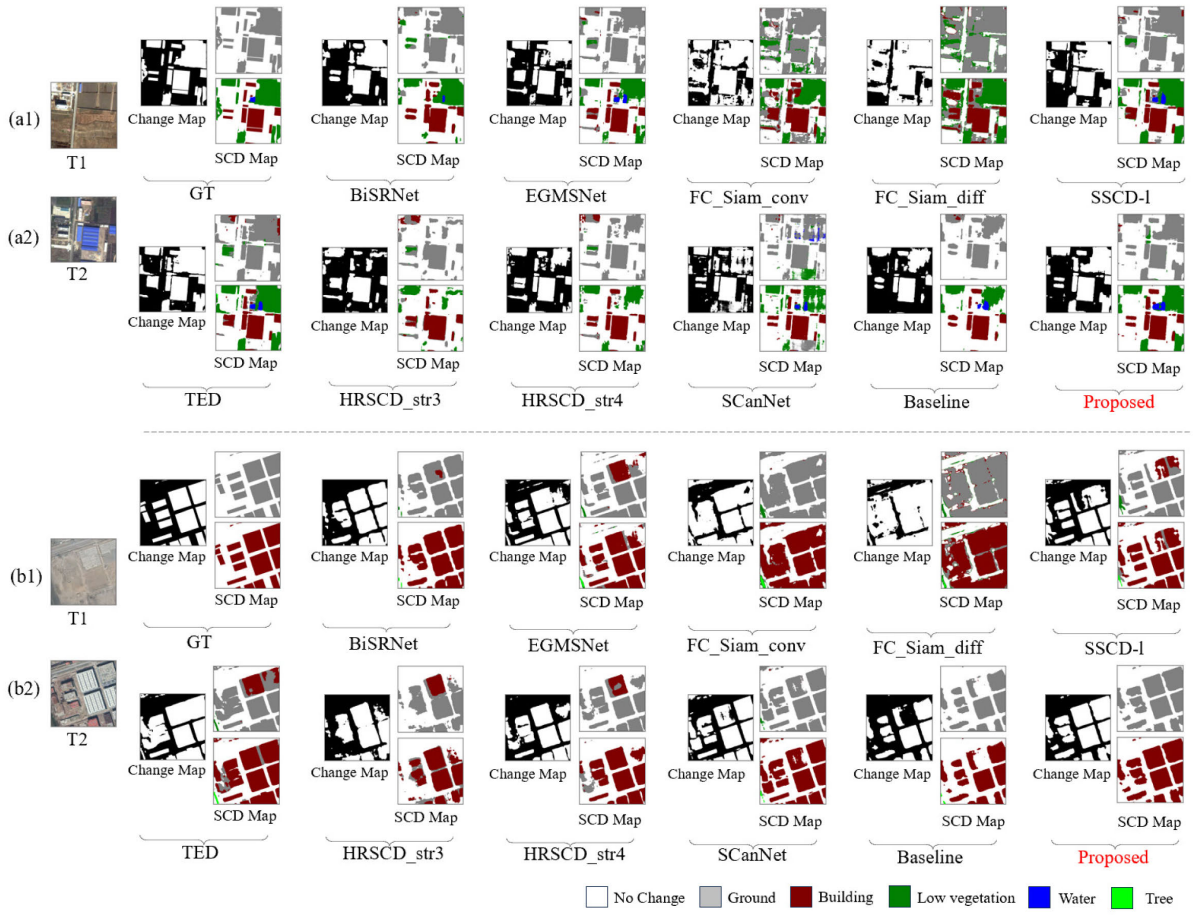


Fig. 5. SCD results among different methods in test areas (a) and (b) on the SECOND dataset.

TABLE I

SCD RESULTS OF THE COMPARATIVE EXPERIMENTS ON THE SECOND DATASET (%). BEST AND SECOND-BEST VALUES ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY. IMPROVEMENT IS REPORTED ONLY WHEN SFSC-CDNET OUTPERFORMS THE SECOND-BEST RESULT; OTHERWISE, "—" IS USED

Type	Method	OA	mIoU	SeK	Fbcd	Fscd	Rec.	Prec.
CNN	BiSRNet [43]	86.72	70.27	18.23	69.60	58.24	**62.71**	54.36
	EGMSNet [44]	86.83	72.79	22.32	<u>73.58</u>	61.57	59.84	63.39
	UNet-3D [45]	79.47	62.11	9.26	61.53	47.69	42.27	52.71
	FC-Siam-conv [20]	80.86	65.80	13.72	66.81	51.99	44.69	62.14
	FC-Siam-diff [20]	79.46	64.68	12.91	65.95	50.01	41.91	61.96
	HRSCD_str3 [46]	83.68	63.75	8.67	60.48	48.18	53.80	43.63
	HRSCD_str4 [46]	85.46	69.78	16.51	68.78	56.11	61.91	51.31
	SSCD-l [43]	86.11	72.17	21.74	73.16	60.76	57.22	<u>64.77</u>
	TED [22]	86.70	72.43	21.20	73.02	60.47	59.65	61.32
Transformer	SCanNet [22]	**87.25**	<u>72.84</u>	<u>22.71</u>	73.41	<u>62.13</u>	61.95	62.32
	ChangeFormerV5 [23]	86.32	70.34	17.55	69.83	57.02	60.32	54.05
	ChangeFormerV6 [23]	86.10	70.16	17.54	69.76	57.13	59.35	55.08
	S-cCDNet [47]	83.89	68.90	15.27	69.16	53.75	50.88	56.96
	SwinSUNet [48]	82.99	66.39	13.08	65.98	52.47	49.77	55.48
Mamba	FEMCD [49]	83.57	67.79	13.46	67.55	52.22	50.34	54.24
	Baseline	87.06	71.87	21.30	72.08	61.35	<u>62.30</u>	60.44
	SFSC-CDNet	<u>87.12</u>	**73.11**	**23.58**	**73.98**	**62.82**	60.84	**64.92**
	Improvement	—	+0.27	+0.87	+0.40	+0.69	—	+0.15

semantic discrimination under class imbalance. Although it does not achieve the highest OA, the simultaneous gains in mIoU and SeK suggest that its advantage lies in the precise delineation of changed regions and their semantic categories,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:06:20 UTC from IEEE Xplore. Restrictions apply.

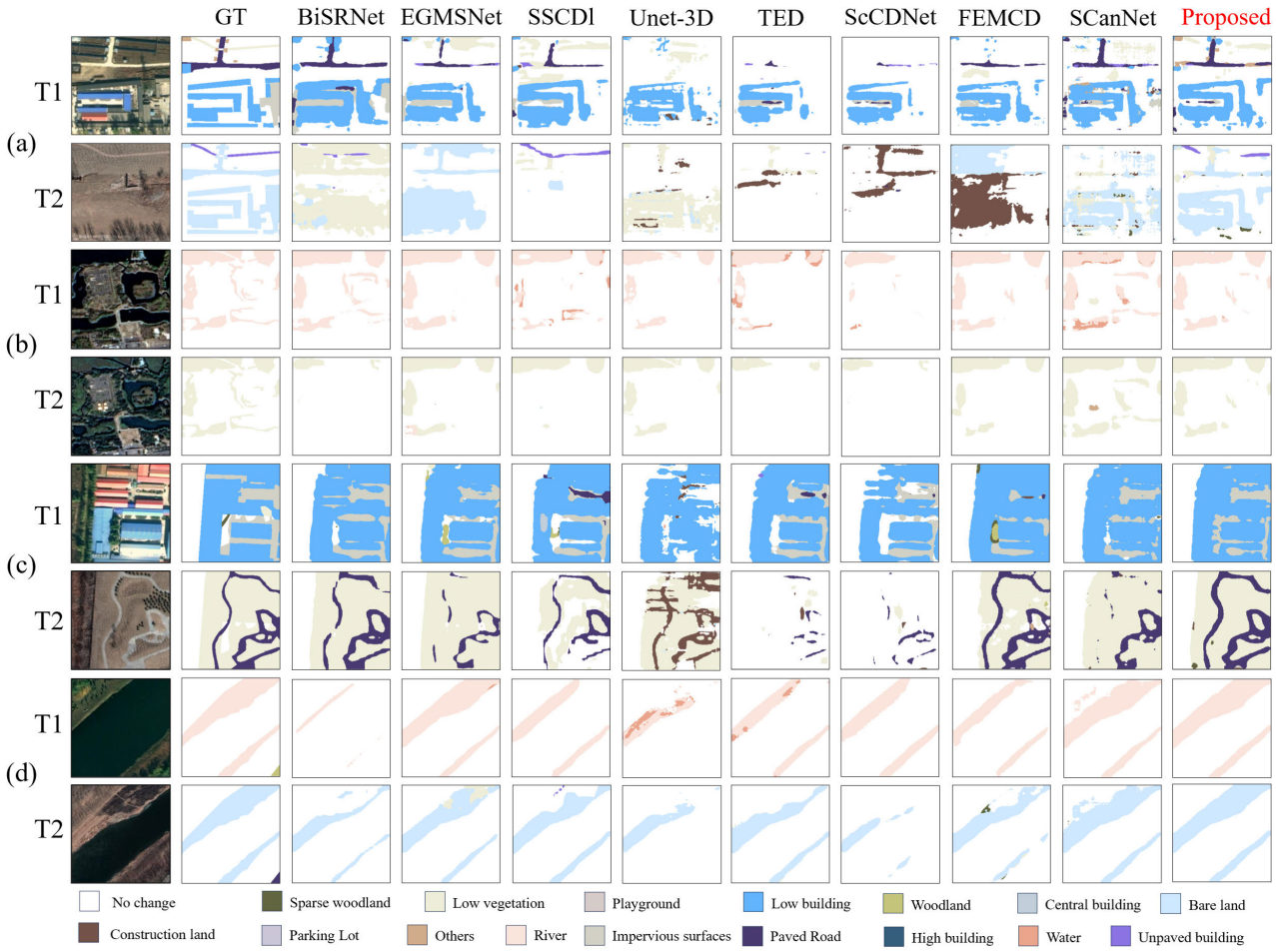


Fig. 6. SCD results among different methods in test areas (a)–(d) on the LevirSCD dataset.

TABLE II

SCD RESULTS OF THE COMPARATIVE EXPERIMENTS ON THE LEVIRSCD DATASET (%). BEST AND SECOND-BEST VALUES ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY. IMPROVEMENT IS REPORTED ONLY WHEN SFSC-CDNET OUTPERFORMS THE SECOND-BEST RESULT; OTHERWISE, "—" IS USED

Type	Method	OA	mIoU	SeK	Fbcd	Fscd	Rec.	Prec.
CNN	BiSRNet [43]	90.47	69.11	13.77	71.26	44.89	58.51	36.42
	EGMSNet [44]	90.32	72.83	15.26	71.00	45.03	51.83	39.81
	UNet-3D [45]	89.51	62.42	6.54	68.16	31.72	56.01	22.12
	FC-Siam-conv [20]	89.05	67.71	4.68	62.69	22.69	39.39	20.67
	FC-Siam-diff [20]	88.62	62.34	2.86	60.89	20.73	37.54	14.32
	SSCD-l [43]	89.80	62.95	9.45	71.69	38.02	**61.11**	27.59
	TED [22]	89.16	59.09	6.59	71.86	30.36	58.70	20.48
Transformer	SCanNet [22]	<u>91.00</u>	<u>73.94</u>	<u>17.77</u>	<u>73.26</u>	<u>47.94</u>	56.70	<u>41.52</u>
	ChangeFormerV5 [23]	89.70	62.60	8.83	70.64	37.32	60.42	27.00
	ChangeFormerV6 [23]	89.27	60.64	7.12	68.91	33.41	57.69	23.51
	S-cCDNet [47]	88.79	56.22	4.98	70.08	25.45	58.64	16.25
	SwinSUNet [48]	89.88	71.22	12.47	68.93	42.15	49.51	36.70
Mamba	FEMCD [49]	89.89	63.79	9.73	71.89	38.58	<u>60.58</u>	28.31
	Baseline	90.03	73.14	17.41	72.51	46.53	54.08	40.83
	SFSC-CDNet	**91.06**	**76.29**	**21.65**	**74.70**	**52.80**	54.97	**50.80**
	Improvement	+0.06	+2.35	+3.88	+1.44	+4.86	—	+9.28

rather than in OA, which is predominantly influenced by unchanged pixels. In comparison to the pure Mamba-encoding baseline, SFSC-CDNet consistently yields improvements of 1.24% in mIoU, 2.28% in SeK, 1.90% in F_{bcd}, and 1.47% in F_{scd}, demonstrating the advantages of richer multitemporal interaction modeling. Qualitative results presented in Fig. 5 corroborate these findings: SFSC-CDNet produces more spatially coherent binary and semantic change maps, suppresses

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:06:20 UTC from IEEE Xplore. Restrictions apply.

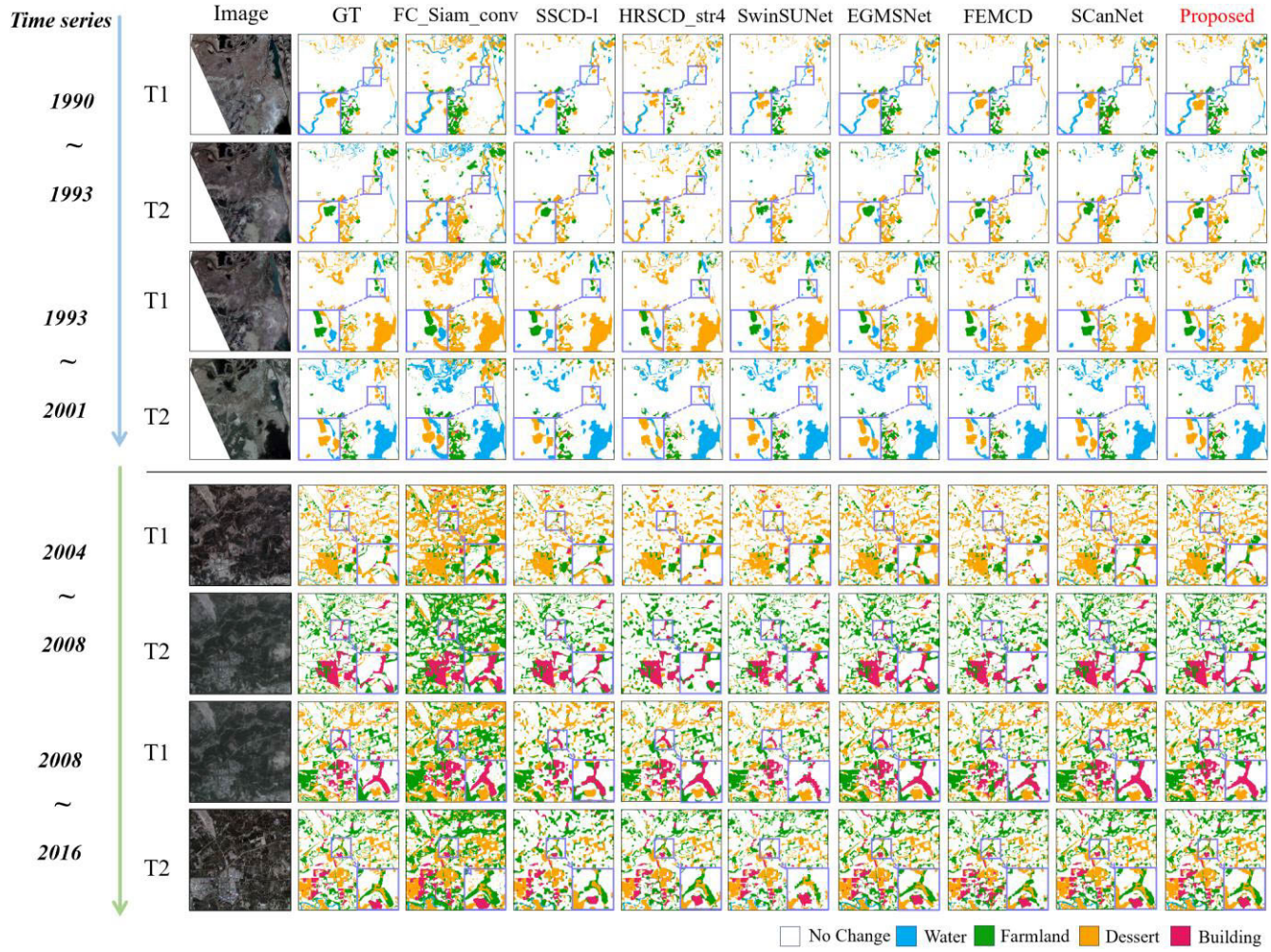


Fig. 7. Time series of SCD results for different methods on the Landsat-SCD dataset in test areas. The purple boxes indicate the same locally enlarged regions across different methods for clearer visual comparison.

scattered noise in complex scenes, and better preserves object boundaries and fine structures.

2) Experiments on LevirSCD: On the LevirSCD dataset, we further assess the performance of SFSC-CDNet as presented in Table II. When compared to the current SOTA model, SCanNet, SFSC-CDNet demonstrates notable improvements in mIoU, SeK, and F_{scd}, with enhancements of 2.35%, 3.88%, and 4.86%, respectively. While maintaining comparable recall, precision experiences a significant increase of 9.28%, suggesting that SFSC-CDNet effectively preserves change coverage while more effectively suppressing false alarms in unchanged regions. As shown in Fig. 6, certain CNN-based methods tend to produce fractures and holes along building boundaries and elongated structures, leading to discontinuous change maps. Although Transformers enhance global consistency, they can still cause semantic drift and over-smoothing in areas characterized by spectral similarity or weak texture. While Mamba-based methods exhibit robust long-range modeling capabilities, the lack of explicit spatial and semantic continuity constraints may lead to scattered noise, loss of boundary structures, or even severe cross-temporal class confusion. In contrast, SFSC-CDNet generates more coherent change regions with improved contour alignment and semantic stability, particularly reducing false detections around fine-grained structures such as roads and the edges of building complexes, which aligns with its superior SeK and precision.

3) Experiments on Landsat-SCD: The Landsat-SCD dataset encompasses extensive areas characterized by coarse-grained land-cover units, including farmland, deserts, water bodies, and urban regions. The observed changes are primarily attributed to cross-regional expansion and gradual evolution, necessitating an evaluation approach that emphasizes not only boundary alignment but also macro-pattern delineation and temporal semantic consistency. As illustrated in Fig. 7, certain methods, such as FC-Siam-conv and ChangeFormerV6, produce fragmented responses in large regions exhibiting subtle texture variations. This fragmentation introduces discrete noise in change maps and results in inconsistent labeling of the same land-cover unit across adjacent time intervals, leading to mismatches between change localization and semantic attribution. In contrast, SFSC-CDNet demonstrates superior temporal semantic continuity across multitemporal comparisons (1990–1993, 1993–2001, 2004–2008, and 2008–2016), where change areas manifest

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:06:20 UTC from IEEE Xplore. Restrictions apply.

TABLE III

SCD RESULTS OF THE COMPARATIVE EXPERIMENTS ON THE LANDSAT-SCD DATASET (%). BEST AND SECOND-BEST VALUES ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY. IMPROVEMENT IS REPORTED ONLY WHEN SFSC-CDNET OUTPERFORMS THE SECOND-BEST RESULT; OTHERWISE, "—" IS USED

Type	Method	OA	mIoU	SeK	Fbcd	Fscd	Rec.	Prec.
CNN	BiSRNet [43]	94.98	86.11	53.15	87.59	86.18	85.85	86.51
	EGMSNet [44]	96.26	89.70	63.39	91.12	90.06	86.47	93.98
	UNet-3D [45]	83.35	67.54	23.50	70.25	66.06	51.96	90.65
	FC-Siam-conv [20]	89.08	75.62	35.05	78.12	76.30	63.31	**95.97**
	FC-Siam-diff [20]	86.91	72.42	29.61	74.93	71.87	59.62	92.84
	HRSCD_str3 [46]	93.26	81.29	41.86	82.53	81.27	83.13	79.50
	HRSCD_str4 [46]	94.12	84.05	47.73	85.52	83.71	85.77	83.95
	SSCD-l [43]	95.07	86.13	53.12	87.54	86.20	87.51	84.94
	TED [22]	96.27	89.35	61.93	90.69	89.57	<u>90.10</u>	89.03
Transformer	SCanNet [22]	<u>96.43</u>	<u>90.04</u>	<u>64.49</u>	<u>91.44</u>	<u>90.53</u>	87.10	<u>94.42</u>
	ChangeFormerV5 [23]	96.26	89.69	62.36	91.02	89.38	89.31	89.46
	ChangeFormerV6 [23]	96.34	89.59	62.55	90.91	89.73	**90.34**	89.13
	S-cCDNet [47]	95.38	86.84	55.39	88.27	87.32	87.82	86.83
	SwinSUNet [48]	95.79	88.06	58.33	89.44	88.28	89.18	87.40
Mamba	FEMCD [49]	94.78	85.30	51.29	86.74	85.56	86.42	84.72
	Baseline	94.45	85.02	57.06	86.67	85.43	82.01	89.15
	SFSC-CDNet	**96.67**	**90.64**	**66.17**	**91.97**	**91.10**	88.39	93.38
	Improvement	+0.24	+0.60	+1.68	+0.53	+0.57	—	—



Fig. 8. SCD results among different methods in test areas (a) and (b) on the WUSU dataset. The red boxes indicate the same local regions across different methods for clearer visual comparison.

as more coherent patches and semantic labels exhibit greater stability before and after changes. This advantage is attributed to enhanced spatial–spectral structural modeling during multitemporal interaction and semantic consistency constraints that suppress pixel-level fluctuations, making the framework more suitable for long-term land-cover evolution in medium-resolution or low-resolution sequences. Quantitatively, SFSC-CDNet achieves the highest OA of 96.67%, mIoU

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:06:20 UTC from IEEE Xplore. Restrictions apply.

TABLE IV

SCD RESULTS OF THE COMPARATIVE EXPERIMENTS ON THE WUSU DATASET (%). BEST AND SECOND-BEST VALUES ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY. IMPROVEMENT IS REPORTED ONLY WHEN SFSC-CDNET OUTPERFORMS THE SECOND-BEST RESULT; OTHERWISE, "—" IS USED

Type	Method	OA	mIoU$_{scd}$	SeK	Fbcd	Fscd	Rec.	Prec.	mIoU$_{lc}$	Kappa
CNN	BiSRNet [43]	93.54	76.24	29.26	73.84	65.95	54.50	83.60	58.01	70.86
	EGMSNet [44]	95.58	81.65	40.47	80.57	75.16	65.25	88.71	63.41	78.47
	UNet-3D [45]	92.36	73.55	23.97	70.30	60.83	49.24	79.65	55.36	66.83
	FC-Siam-conv [20]	88.59	66.75	13.64	61.12	47.51	35.77	70.72	40.03	56.24
	FC-Siam-diff [20]	87.88	65.54	8.62	58.92	40.70	31.20	58.54	30.07	53.87
	HRSCD_str3 [46]	90.81	71.50	12.38	66.98	44.45	37.31	54.96	29.07	63.32
	HRSCD_str4 [46]	90.03	71.09	13.63	66.86	44.30	35.37	59.25	29.20	62.97
	SSCD-l [43]	93.76	76.76	30.17	74.46	66.46	55.56	82.80	57.63	71.58
	TED [22]	94.23	78.11	32.22	76.24	68.50	57.29	85.17	57.28	73.59
Transformer	SCanNet [22]	95.32	80.44	38.16	79.07	74.25	64.35	87.85	64.62	76.80
	ChangeFormerV5 [23]	94.09	78.83	30.22	77.09	64.79	55.16	78.51	50.24	74.58
	ChangeFormerV6 [23]	93.90	79.18	29.34	77.57	62.87	53.24	76.76	47.76	75.10
	S-cCDNet [47]	79.52	55.92	8.85	47.25	29.94	20.10	58.72	31.47	39.62
	SwinSUNet [48]	91.87	75.90	17.51	73.22	47.91	40.74	58.20	35.90	70.25
Mamba	FEMCD [49]	94.37	78.83	33.01	77.13	68.55	58.00	83.88	58.72	74.58
	Baseline	<u>95.97</u>	<u>82.35</u>	<u>43.67</u>	<u>81.45</u>	<u>78.35</u>	<u>67.84</u>	<u>92.82</u>	<u>68.78</u>	<u>79.44</u>
	SFSC-CDNet	**96.39**	**83.59**	**47.20**	**82.93**	**80.85**	**70.32**	**95.17**	**70.79**	**81.09**
	Improvement	+0.42	+1.24	+3.53	+1.48	+2.50	+2.48	+2.35	+2.01	+1.65

TABLE V

ABLATION RESULTS FOR THE PROPOSED TECHNIQUES ON THE SECOND DATASET (%). BEST VALUES ARE HIGHLIGHTED IN BOLD

Method	Proposed Techniques				Overall Performance Metrics				
	SFSDI	SCCM	EGIST	$\mathcal{L}_{cwsc}$	OA	mIoU	SeK	Fscd	Fbcd
Proposed					87.06	71.87	21.30	61.35	72.08
	√				86.89	71.94	21.87	61.81	72.20
	√	√			87.31	72.68	22.75	62.49	73.06
	√	√	√		**87.58**	73.02	23.28	62.68	73.56
	√	√	√	√	87.12	**73.11**	**23.58**	**62.82**	**73.98**

of 90.64%, SeK of 66.17%, F_{scd} of 91.10%, and F_{bcd} of 91.97 (Table III).

4) Experiments on WUSU: In multitemporal SCD, the evolution of LCLU from T1 to T2 to T3 is typically phased and cumulative. Such phased evolution places higher demands on maintaining temporal consistency and spatial coherence in semantic representations under multiphase observations, while accurately inferring the target first-to-last semantic change with the support of intermediate semantic states. As shown in Table IV, the metrics mIoU$_{scd}$, SeK, F_{scd}, and F_{bcd} improve by 1.24%, 3.53%, 2.50%, and 1.48% over the second-best methods, respectively. In addition, SFSC-CDNet achieves the highest mIoU$_{lc}$ of 70.79% and Kappa of 81.09%, further demonstrating its robustness in change localization. The multitemporal visualizations in Fig. 8 reveal that several comparative methods exhibit temporal inconsistency in areas undergoing cross-phase evolution: TED tends to generate fragmented predictions, whereas SCanNet and EGMSNet demonstrate improved performance, although category adhesion remains evident. In contrast, SFSC-CDNet produces a clearer evolutionary chain across successive phases, with its change responses aligning more closely with the phased accumulation pattern. The clearer evolutionary pattern indicates that the supervised intermediate semantic state helps constrain the shared representation space, thereby improving the stability of first-to-last semantic change inference.

E. Ablation Experiment

1) Effectiveness of the Proposed Modules: To validate the effectiveness of each key component, we utilize the baseline model without any newly introduced modules as a reference and progressively add SFSDI, SCCM, EGIST, and L$_{cwsc}$ in a stepwise manner (Table V). Overall, the removal of any module degrades performance. The addition of SFSDI consistently improves SeK and F_{scd} from 21.30% to 21.87% and from 61.35% to 61.81%, respectively, indicating a stronger cross-temporal feature interaction. The further introduction of the SCCM decoder yields larger gains, with SeK increasing from 21.87% to 22.75% and F_{bcd} increasing from 72.20% to 73.06%, suggesting improved decoding of change localization and semantic assignment while reducing inconsistency between these aspects. The addition of EGIST further elevates mIoU and SeK from 72.68% to 73.02% and from 22.75% to 23.28%, respectively, reflecting enhanced discriminability of high-level semantics. Finally, incorporating L_{cwsc} achieves the best overall performance for the full model. In

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:06:20 UTC from IEEE Xplore. Restrictions apply.

TABLE VI

ABLATION RESULTS OF WEIGHTED-CONSTRAINT COMBINATIONS IN $\mathcal{L}_{\text{Cwsc}}$ ON THE WUSU DATASET (%). BEST VALUES ARE HIGHLIGHTED IN BOLD

Method	Proposed Techniques				Overall Performance Metrics				
	cwsc_weight	cwsc_same	cwsc_diff	cwsc_dilate	OA	mIoU	SeK	Fscd	Fbcd
G0	0.0	0.0	0.0	0	96.13	82.85	45.70	79.25	82.05
G1	0.3	0.2	0.1	0	**96.39**	**83.59**	**47.20**	**80.85**	**82.93**
G2	0.3	0.3	0.0	0	96.08	82.64	45.38	79.19	81.86
G3	0.3	0.0	0.3	0	96.22	82.77	46.07	79.41	81.96
G4	0.3	0.2	0.1	5	96.23	83.34	46.29	79.83	82.12

TABLE VII

SENSITIVITY ANALYSIS RESULTS FOR $\mathcal{L}_{\text{Cwsc}}$ LOSS WEIGHTS ON THE WUSU DATASET (%). BEST VALUES ARE HIGHLIGHTED IN BOLD

Loss_weight	Overall Performance Metrics					Boundary-aware Metrics	
	OA ↑	mIoU ↑	SeK ↑	Fscd ↑	Fbcd ↑	$F_{boundary}$ ↑	Brier Score ↓
0.1	96.15	82.92	45.19	79.40	82.13	**90.95**	0.96
0.3	**96.39**	**83.59**	**47.20**	**80.85**	**82.93**	90.27	**0.93**
0.5	96.24	83.32	46.35	79.92	82.06	90.54	0.94
0.6	96.15	82.90	45.26	79.44	82.12	90.90	0.96
0.8	96.06	82.64	44.24	78.67	81.79	89.53	1.05

TABLE VIII

RESULTS FOR DIFFERENT FREQUENCY-DOMAIN COMPONENTS ON THE LEVIRSCD DATASET (%). BEST VALUES ARE HIGHLIGHTED IN BOLD

Method	Overall Performance Metrics					Boundary-aware Metrics	
	OA ↑	mIoU ↑	SeK ↑	Fscd ↑	Fbcd ↑	$F_{boundary}$ ↑	Brier Score ↓
LF	91.34	75.62	**21.58**	**53.30**	**74.04**	39.75	**5.07**
HF	**91.36**	75.30	21.29	52.78	73.67	**42.71**	5.19
BL	90.76	**75.85**	20.95	51.61	73.71	36.06	5.27
BR	91.12	74.81	19.24	49.51	73.36	36.06	5.12

summary, these components provide complementary benefits that enhance change-detection reliability and stabilize semantic change discrimination.

2) Effectiveness of the $\mathcal{L}_{Cwsc}$-Guided Consistency Learning: To analyze the contribution of each subterm in $\mathcal{L}_{cwsc}$, we ablate its weighted components, including cwsc_weight, cwsc_same, cwsc_diff, and cwsc_dilate (Table VI, G0–G4). The incorporation of $\mathcal{L}_{cwsc}$ yields more stable optimization for both semantic discrimination in change regions and binary change localization. Among all variants, G1 achieves the highest overall balance, with OA of 96.39%, mIoU of 83.59%, SeK of 47.20%, F_{scd} of 80.85%, and F_{bcd} of 82.93%. This advantage arises because cwsc_same enforces semantic consistency in unchanged areas to reduce temporal drift, while cwsc_diff promotes semantic separability in changed areas to sharpen class-transition boundaries. Furthermore, cwsc_weight reallocates gradients to critical regions, alleviating optimization bias caused by class imbalance and sparse changes. In contrast, G2 and G3 overemphasize either the same or the difference term, disrupting the balance between consistency and separability, which results in overly conservative or unstable semantic responses. Although a larger dilation in G4 increases fault tolerance by expanding supervision, it also introduces boundary-mixed pixels into strict constraints, leading to noisy supervision and weakening the overall tradeoff.

3) Effectiveness and Sensitivity Analysis of $\mathcal{L}_{Cwsc}$ Loss Weights: The impact of the $\mathcal{L}_{cwsc}$ weight on training stability and performance is analyzed through sensitivity and ablation experiments, maintaining all other settings fixed constant while varying the loss weight from 0.1 to 0.8 (Table VII). The results exhibit a nonmonotonic trend, revealing a distinct optimum. At a weight of 0.3, the metrics mIoU, SeK, F_{scd}, and F_{bcd} achieve optimal or near-optimal values, while the Brier score is minimized, suggesting an ideal balance between the cross-temporal consistency constraint and the task discrimination objective. When the weight is increased to 0.5 or higher, overall performance deteriorates, suggesting that an excessively strong consistency constraint can dominate the optimization process and induce over-smoothing, thereby impairing change discrimination. Conversely, a weight that is too small (e.g., 0.1) provides insufficient regularization to stabilize cross-temporal semantic mapping, resulting in limited gains. Consequently, the optimal default weight of $\mathcal{L}_{cwsc}$ is recommended to be set at 0.3.

V. DISCUSSION

A. Superiority of the SFSC-CDNet

1) Frequency-Aware Evaluation and Correlation Analysis: The significance of frequency-domain cues was examined through a comparative analysis of four components, integrating

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:06:20 UTC from IEEE Xplore. Restrictions apply.

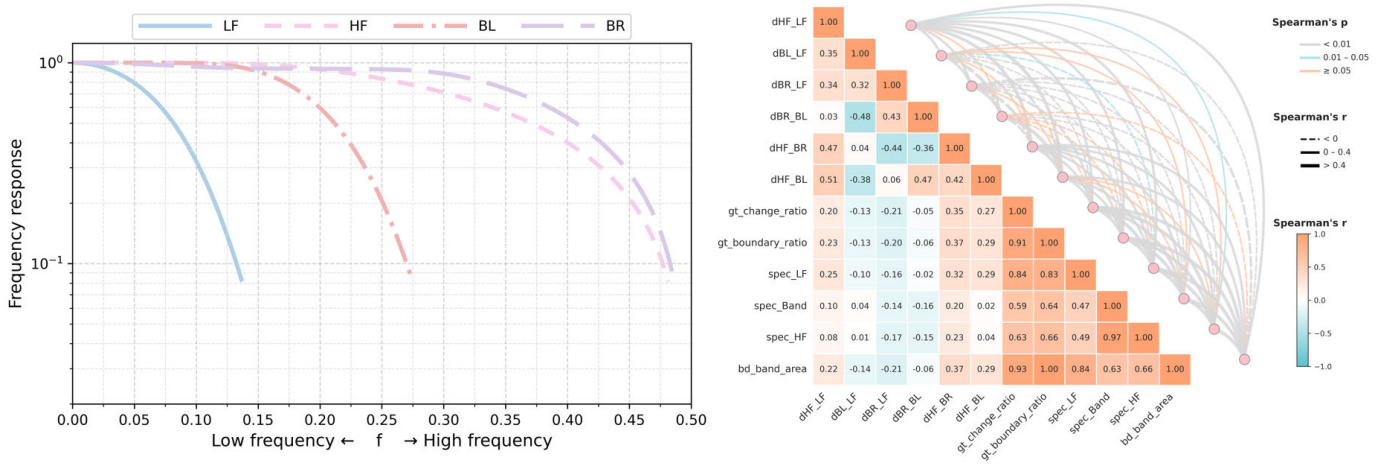


Fig. 9. Distributions of frequency-domain components and Spearman correlation analysis.

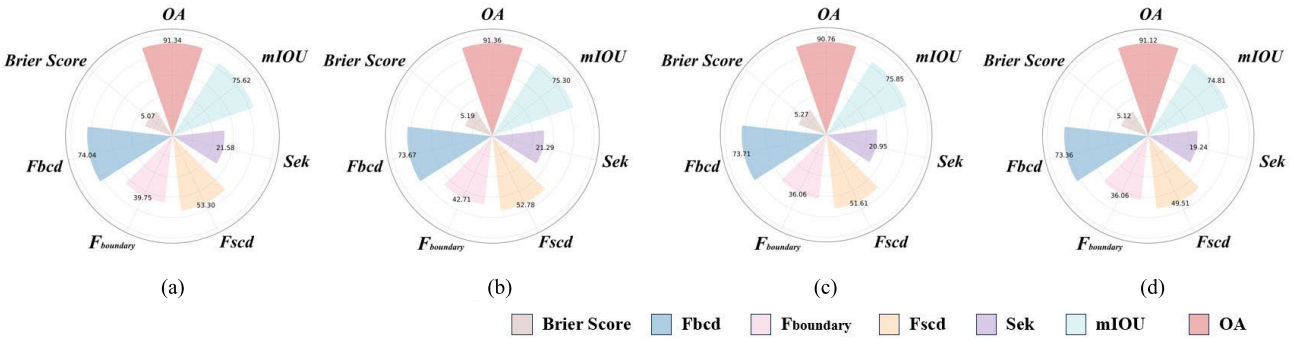


Fig. 10. Results of using different frequency-domain components on the LevirSCD dataset. (a) LF, (b) HF, (c) BL, and (d) BR.

TABLE IX
RESULTS FOR DIFFERENT DISCREPANCY MODELING MEASURES ON THE LEVIRSCD DATASET (%). BEST VALUES ARE HIGHLIGHTED IN BOLD

Method	Overall Performance Metrics					Boundary-aware Metrics	
	OA ↑	mIoU ↑	SeK ↑	Fscd ↑	Fbcd ↑	$F_{boundary}$ ↑	Brier Score ↓
M0	90.81	75.84	20.26	51.61	74.04	39.50	5.28
M1	91.06	**76.29**	**21.65**	**52.80**	**74.70**	42.60	5.24
M2	91.06	75.90	20.90	51.26	74.41	**43.10**	5.15
M3	**91.30**	75.76	21.32	52.38	74.28	41.67	**5.05**

frequency-response characteristics with Spearman correlation analysis. Fig. 9 illustrates that low-frequency (LF) components concentrate on lower frequencies with rapid attenuation, favoring regional structure and semantic contours. In contrast, HF and bandpass (BR) components extend further into higher frequencies, emphasizing edges and fine textures, while band-limited (BL) components occupy an intermediate position. The correlation analysis reveals two key findings. First, the proportion of change exhibits a strong correlation with the boundary proportion, indicating that larger change areas typically yield more complex boundaries. Second, HF or bandpass statistics are closely aligned with boundary complexity: the correlations between HF and BR are notably high, and these metrics are also strongly correlated with the change proportion (ρ = 0.93). Consistently, HF difference terms (e.g., HF and BR) demonstrate higher correlations with the boundary complexity metric (0.35/0.37 for dHF_BR) compared to the LF term dHF_LF (0.20/0.23). To mitigate bias from a single metric, we further present multimetric evaluations in Fig. 10 and Table VIII. The results indicate that LF terms achieve the most stable overall semantic change discrimination, supporting their role in mitigating semantic drift caused by imaging discrepancies. Although HF terms perform best on boundary-related metrics, they slightly degrade overall performance and Brier score, suggesting that increased contour sensitivity may introduce texture-induced false changes. Conversely, BL and BR yield weaker overall results, indicating that partial spectral representations are adequate for jointly preserving regional coherence and semantic consistency. In summary, LF cues primarily support macro-level semantic stability, while HF

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:06:20 UTC from IEEE Xplore. Restrictions apply.

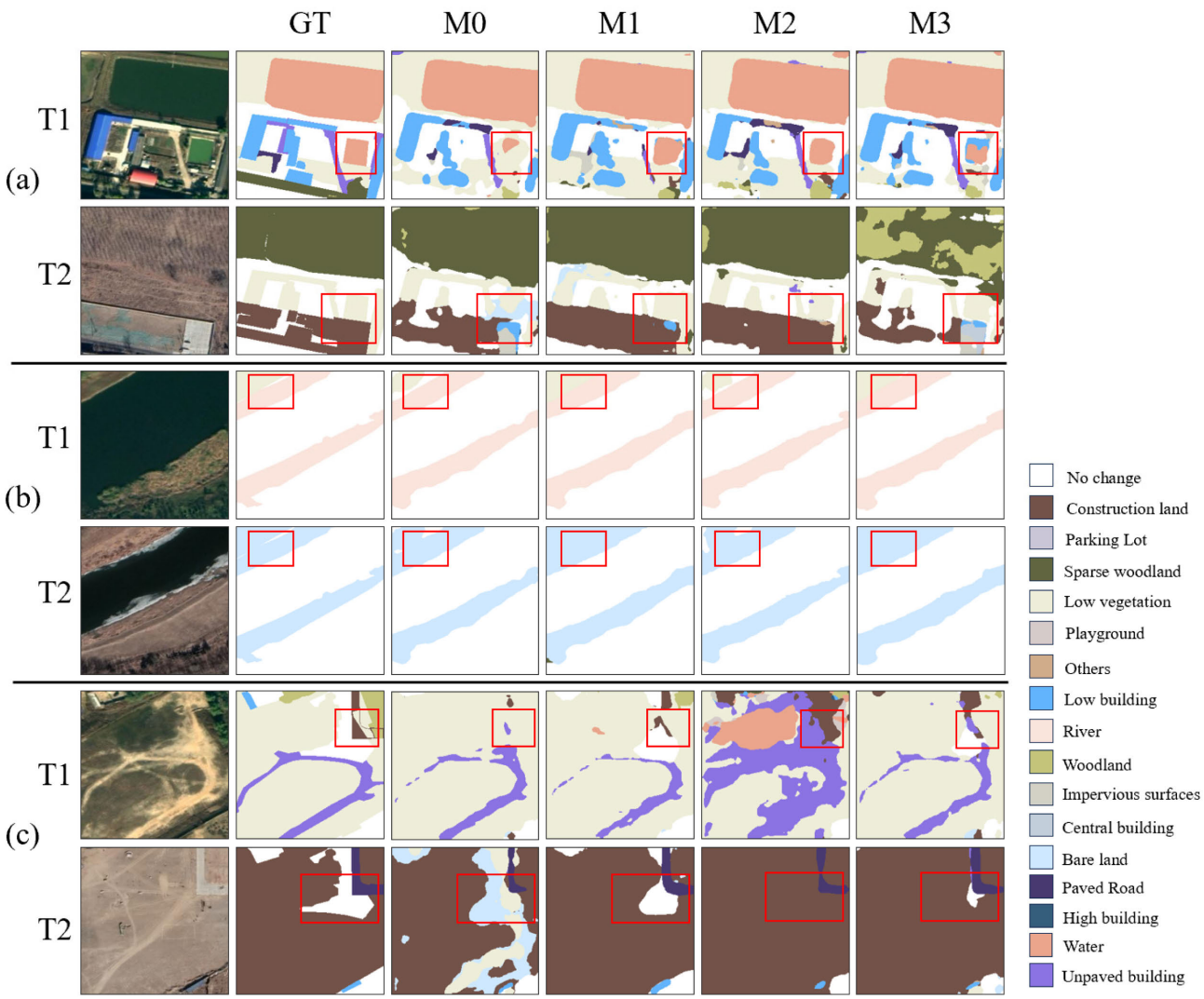


Fig. 11. SCD results with different discrepancy measures in test areas (a)–(c) on the LevirSCD dataset.

cues predominantly capture boundary detail. Their combination is essential for balancing false-change suppression and boundary fidelity. These results show that the prompt-guided spectral attention in SFSDI adaptively adjusts the contributions of low- and HF responses. It retains the stable semantic information reflected by LF cues and strengthens the boundary details captured by HF cues, thereby improving cross-temporal difference representation.

2) Effectiveness of the EGIST-Guided Discrepancy Modeling: To identify the change cues that drive gating and edge enhancement in EGIST, we exclusively replaced the cross-temporal difference metric $\Delta(\cdot)$, allowing performance differences to be attributed solely to the metric itself. The four metrics range from amplitude sensitivity to structural robustness. M0, the L1 energy difference, emphasizes element-wise amplitude changes, which sharpens edges but is highly sensitive to radiometric or spectral perturbations. In contrast, M2 (cosine similarity) and M3 (mean-centered correlation) focus on directional or structural coherence, enhancing robustness to scaling and global shifts while potentially weakening local HF boundary responses. Results presented in Table IX indicate that M1, the L2 energy difference, offers the best overall balance. Compared to M0, it improves OA, SeK, F_{scd}, and F_{bcd}, increases $F_{boundary}$ from 39.50% to 42.60%, and maintains a low Brier score of 5.24, reflecting better boundary consistency with reliable calibration. Although M2 further raises the metric $F_{boundary}$ to 43.10%, it decreases F_{scd} from 52.80% to 51.26%, suggesting that overly conservative semantic discrimination may occur when directional consistency prevails. M3 achieves the lowest Brier score at 5.05; however, its overall and boundary metrics remain inferior to M1, indicating that strict mean-centered correlation may compromise boundary clarity. As illustrated in Fig. 11, amplitude-based metrics tend to introduce local spurious responses, while directional-correlation metrics oversmooth details; M1 best preserves the balance between change coherence and boundary separability.

3) Feature Visualization of SCCM-Guided Semantic Decoding: Intermediate feature responses are visualized in Fig. 12, with red indicating high activation, which validates the effect of semantic decoding at both the encoder stage and the subsequent SCCM decoder. The encoder features respond to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:06:20 UTC from IEEE Xplore. Restrictions apply.

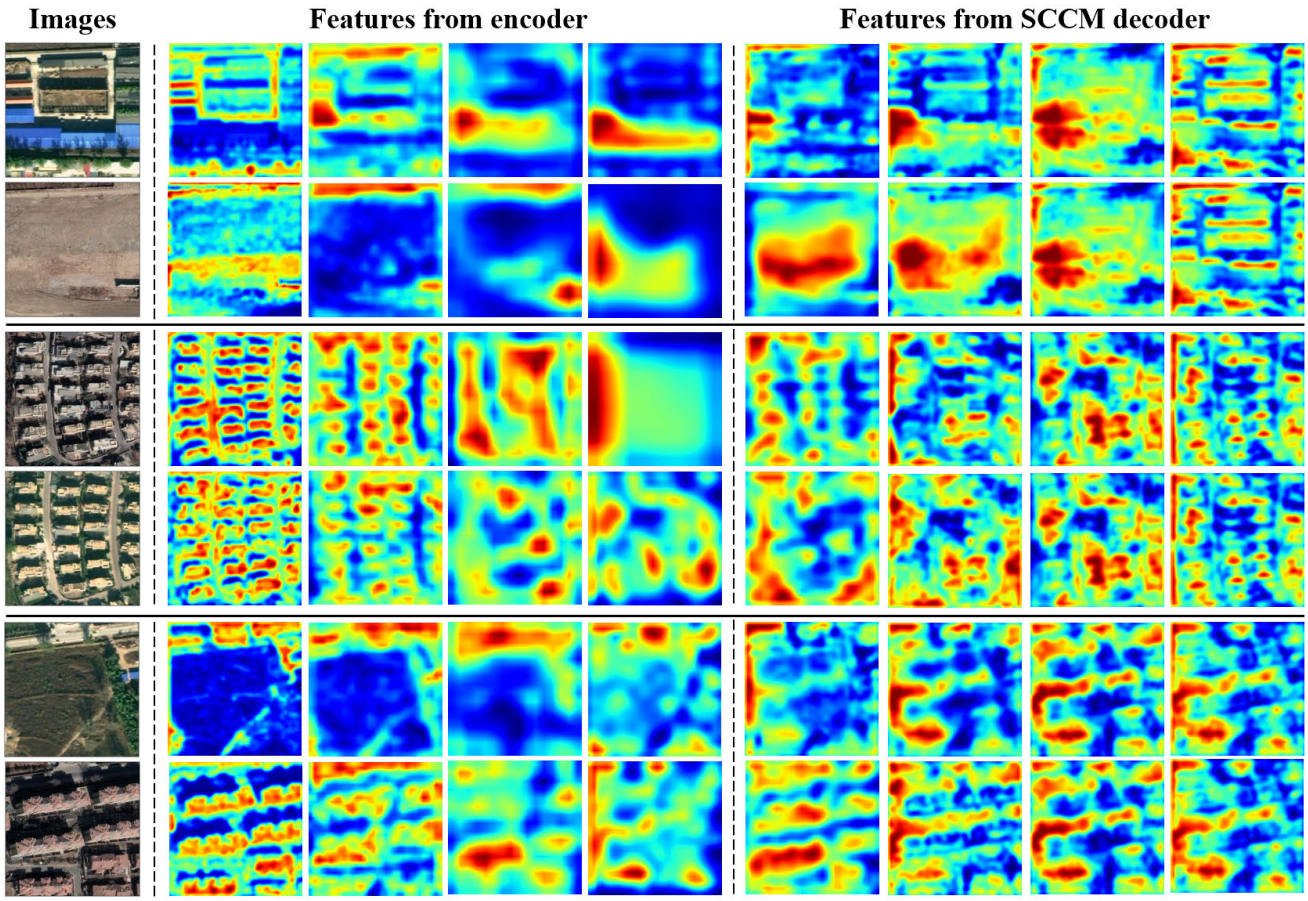


Fig. 12. Visualization of feature maps extracted from the encoder and the SCCM decoder. Red and blue regions denote higher and lower activation intensities, respectively. The SCCM decoder produces more concentrated responses in semantically relevant regions.

targets and textures; however, in regions characterized by significant temporal discrepancies, complex backgrounds, or similar interclass appearances, activations tend to disperse and extend into unchanged areas. This leads to inconsistent cues between change localization and semantic segmentation. After SCCM decoding, high-response regions become more concentrated on structural boundaries and altered semantic objects, while background noise and spurious change textures are effectively suppressed. The resulting high-level features continue to preserve essential textures and structural outlines, demonstrating clearer and more focused responses within true change regions.

4) Semantic Change Transition Matrix Analysis: To further evaluate the model's capability in capturing semantic change patterns from a category-transfer perspective, we computed pixel-level statistics on the predictions and visualized intertemporal transfers using a chord diagram. The arc segments in Fig. 13 indicate category proportions, while the chords represent transfer flows from T1 to T2. Compared to the ground truth (GT), most comparative methods (e.g., SSCD-l) exhibit multisource divergence. Beyond the dominant transfer paths, numerous weak and scattered cross-category edges persist, implying a higher risk of semantic drift in change regions and boundary neighborhoods, particularly between spectrally and texturally similar classes, such as low/high buildings and low vegetation/forest. In contrast, SFSC-CDNet yields a chord structure that is closer to GT: self-loops of unchanged categories are more concentrated, indicating stronger semantic retention in stable areas. In addition, change edges are more focused on a few dominant transfers, thereby suppressing scattered false transfers and long-tail edges. Overall, SFSC-CDNet better aligns with the category-transfer distribution implied by GT.

B. Comparison of Backbone Encoder Types and Temporal Consistency Analysis

Backbone replaceability is crucial for assessing the architectural generalization of SFSC-CDNet. Therefore, we adopted three representative backbones: CNN (ResNet18/34/50) [50], Transformer (SegFormer-b0–b5) [51], and Mamba (GOST-Mamba-tiny/small) [52]. The results presented in Table X demonstrate that SFSC-CDNet consistently achieves performance gains across all backbone families, indicating strong transferability. When considering parameter count (Params) and computational complexity (GFLOPs), clear performance–efficiency tradeoffs can be observed. CNN-based backbones generally provide higher accuracy at the expense of substantially increased computational costs, whereas Transformer-based backbones, such as b0, reduce computational complexity with relatively limited accuracy. Scaling

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:06:20 UTC from IEEE Xplore. Restrictions apply.

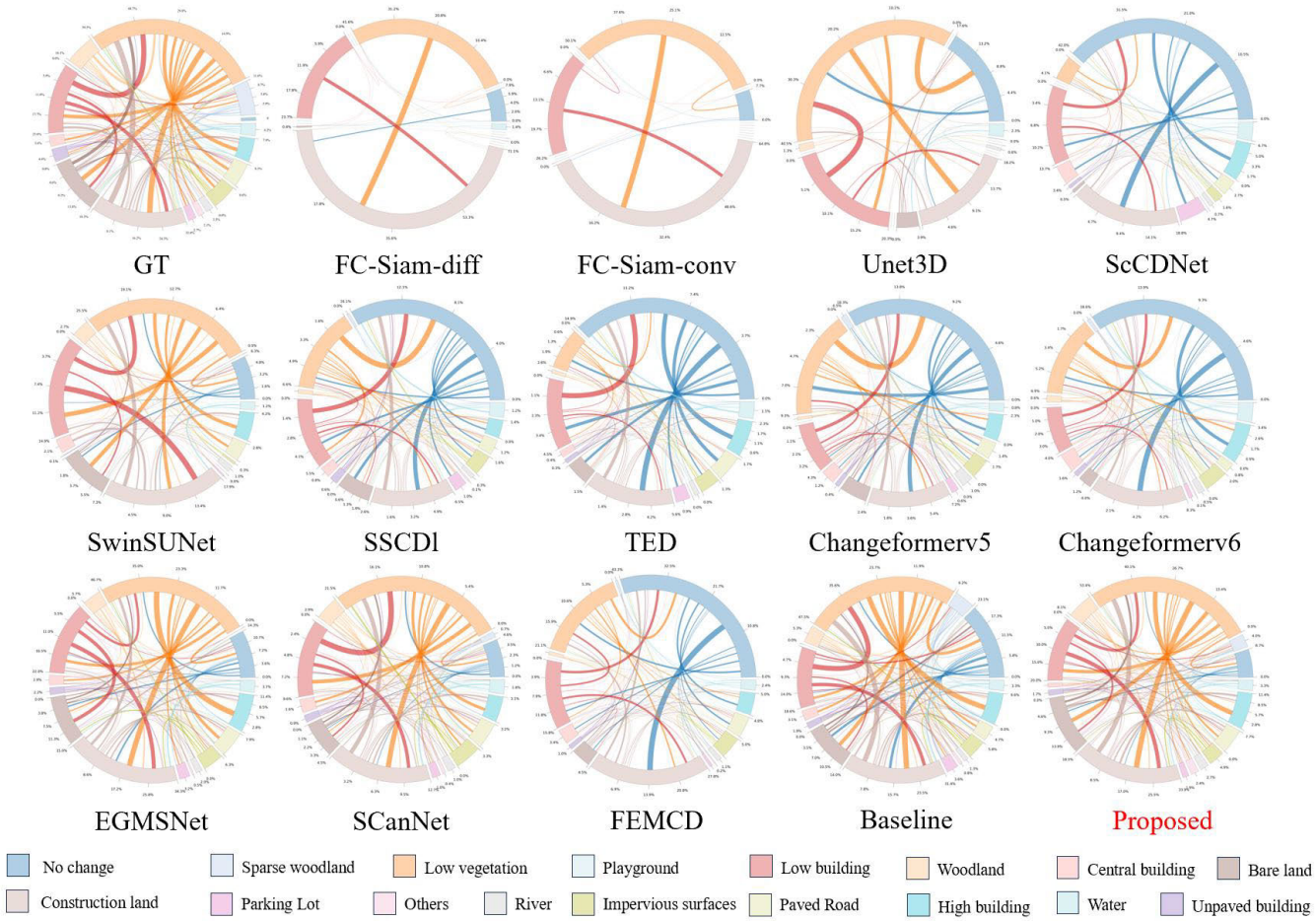


Fig. 13. Comparison of semantic class transitions for different methods on the LevirSCD dataset.

TABLE X

COMPARISON RESULTS OF DIFFERENT BACKBONE ENCODERS ON THE LANDSAT-SCD DATASET (%). BEST VALUES ARE HIGHLIGHTED IN BOLD

Type	Backbone	OA ↑	mIoU ↑	SeK ↑	Fbcd ↑	Fscd ↑	Params/M ↓	GFLOPs ↓
CNN	ResNet18 [50]	94.84	86.39	54.04	88.03	86.27	13.21	146.44
	ResNet34 [50]	95.44	87.87	57.78	89.43	87.74	23.32	258.26
	ResNet50 [50]	95.57	88.40	59.19	89.97	88.13	33.75	356.26
Transformer	SegFormer-b0 [51]	93.40	83.17	46.25	84.91	82.73	**7.72**	73.62
	SegFormer-b1 [51]	94.58	85.65	52.50	87.33	85.82	20.85	94.35
	SegFormer-b2 [51]	95.35	87.39	57.17	88.99	87.84	31.89	115.52
	SegFormer-b3 [51]	95.55	87.80	58.38	89.37	88.39	51.77	154.20
	SegFormer-b4 [51]	95.89	88.65	60.64	90.16	89.22	65.94	191.34
	SegFormer-b5 [51]	95.83	88.48	60.17	90.01	89.06	89.15	229.18
Mamba	GOST-Mamba-tiny [52]	96.53	90.27	65.05	91.21	90.71	39.90	**67.42**
	GOST-Mamba-small [52]	**96.67**	**90.64**	**66.17**	**91.97**	**91.10**	79.64	72.84

from b3 to b5 yields performance comparable to strong CNN-based backbones, accompanied by a rapid increase in parameter count, reaching up to 89.15 M. In contrast, GOST-Mamba provides a more favorable balance: GOST-Mamba-tiny achieves leading metrics with low GFLOPs, whereas GOST-Mamba-small attains the best accuracy with relatively low computational cost (72.84 GFLOPs, comparable to Transformer-based backbones) and a parameter count of 79.64 M. Overall, SFSC-CDNet demonstrates robust generalization across backbone paradigms, with Mamba being particularly effective in balancing global modeling and efficiency.

The continuous temporal visualization depicted in Fig. 14 suggests that the Mamba backbone yields greater stability in temporal consistency. It effectively preserves semantic consistency in unchanged regions and mitigates temporal drift when changes between adjacent years are minimal. In continuously evolving areas, it produces coherent incremental category

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:06:20 UTC from IEEE Xplore. Restrictions apply.

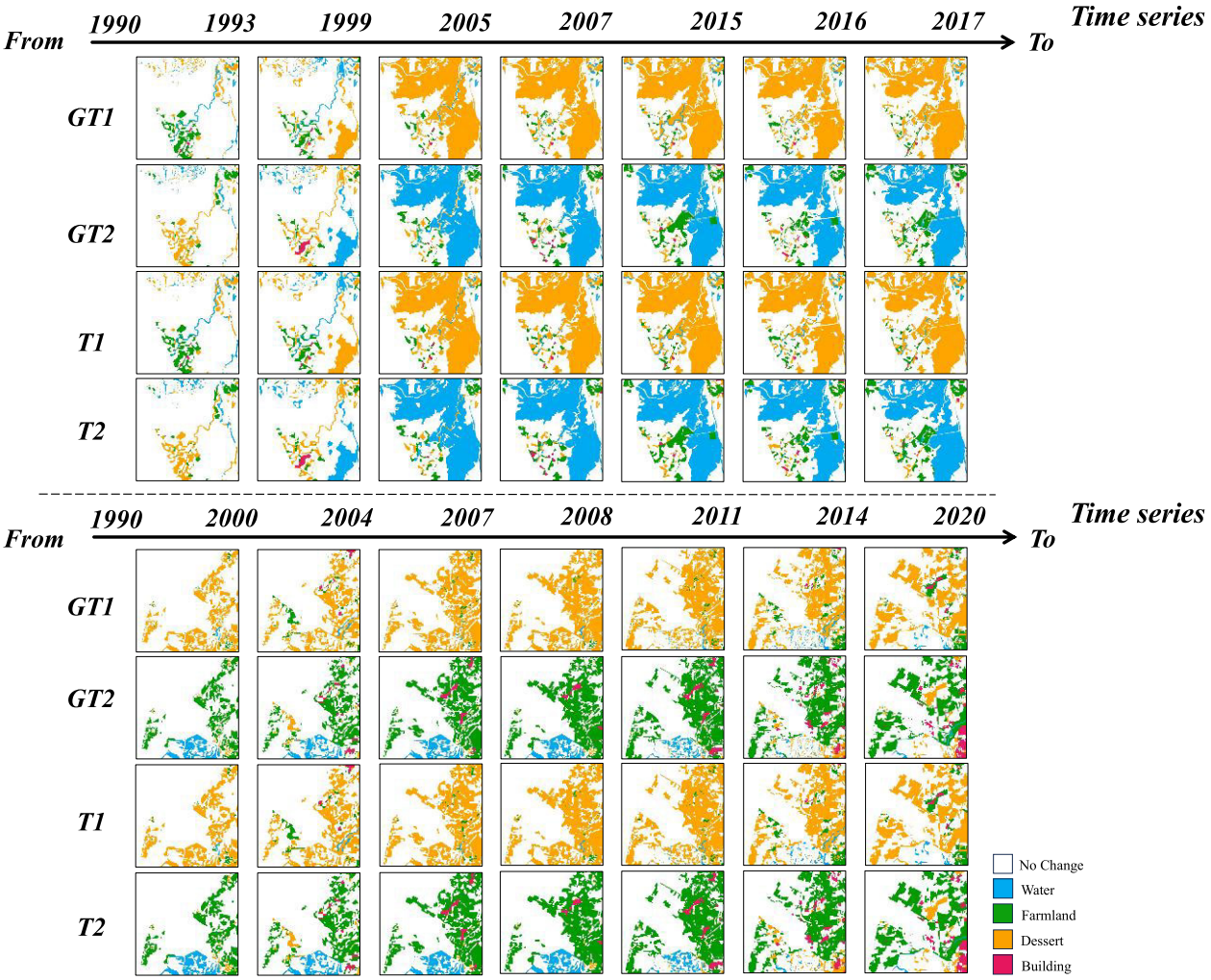


Fig. 14. Time-series visualization of SCD results with the Mamba-based backbone encoder on the Landsat-SCD dataset.

TABLE XI
COMPARISON RESULTS OF DIFFERENT INPUT SIZES ON THE WUSU DATASET (%). BEST VALUES ARE HIGHLIGHTED IN BOLD

Input Size	Overall Performance Metrics					Boundary-aware Metrics	
	OA ↑	mIoU ↑	SeK ↑	Fscd ↑	Fbcd ↑	$F_{boundary}$ ↑	Brier Score ↓
64	93.65	76.58	26.26	62.44	74.01	**95.15**	2.69
128	94.97	79.89	35.83	71.63	78.39	93.79	1.76
256	95.85	82.09	42.48	77.31	81.12	91.86	1.20
384	96.11	82.76	45.03	79.38	81.95	90.63	0.98
512	**96.39**	**83.59**	**47.20**	**80.85**	**82.93**	90.27	**0.93**

updates rather than abrupt shifts. This indicates that the capacity for modeling temporal dependencies directly influences the quality of cross-temporal alignment and the interpretability of semantic evolution trajectories. With enhanced long-range modeling capabilities, Mamba can more effectively leverage multitemporal cues to suppress noise and maintain stable semantic transitions. The selection of the backbone encoder substantially influences the feature representation capability and computational cost of the overall framework. Nevertheless, SFSDI, EGIST, and SCCM are designed as task-specific modules for cross-temporal difference modeling, structure-guided feature interaction, and semantic consistency learning, respectively, and are therefore not inherently restricted to a specific encoder family. The superior results obtained with GOST-Mamba suggest that its temporal and contextual modeling capability is more compatible with the proposed modules. Therefore, the performance gains of SFSC-CDNet should be attributed to the synergy between backbone representation capacity and the proposed task-specific designs, rather than to the backbone alone.

C. Impact of Multiscale Input Settings

This section further assesses the influence of input size on boundary detection. As presented in Table XI, an increase

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:06:20 UTC from IEEE Xplore. Restrictions apply.

TABLE XII
COMPARISON RESULTS OF COMPUTATIONAL EFFICIENCY AMONG DIFFERENT METHODS

Type	Method	Params/M	GFLOPs
CNN	BiSRNet [43]	23.39	190.16
	EGMSNet [44]	23.92	216.09
	UNet-3D [45]	6.16	1239.28
	FC-Siam-conv [20]	2.74	35.12
	FC-Siam-diff [20]	1.66	21.52
	HRSCD_str3 [46]	11.64	32.75
	HRSCD_str4 [46]	11.86	33.60
	SSCD-l [43]	23.31	189.76
	TED [22]	24.19	204.29
Transformer	SCanNet [22]	27.90	264.95
	ChangeFormerV5 [23]	60.27	580.86
	ChangeFormerV6 [23]	46.03	572.39
	S-cCDNet [47]	8.99	61.08
	SwinSUNet [48]	28.75	41.12
Mamba	FEMCD [49]	56.84	120.42
	SFSC-CDNet-tiny	39.90	67.42
	SFSC-CDNet-small	79.64	72.84

in input size correlates with continuous improvement across overall metrics, including OA, mIoU, SeK, F_{scd}, and F_{bcd}. Conversely, $F_{boundary}$ exhibits a decline as the input size increases. This phenomenon primarily arises from the strong correlation between boundary metrics and local pixel-level details. When dealing with smaller inputs, the target and boundary tend to form a relatively sharp transition following downsampling, prompting the model to generate high-contrast edge responses and thereby achieving a higher boundary matching rate within the tolerance bandwidth τ. In contrast, with larger inputs, the prediction results prioritize semantic continuity, leading to more thorough preservation of mixed pixels and fine-grained textures near the boundary. This results in a boundary morphology that is more realistic yet complex (e.g., featuring more fragmented contours or multipixel transition zones), which, under a fixed τ setting, consequently reduces the $F_{boundary}$.

D. Model Parameters and Computational Complexity

To systematically evaluate computational costs, we report the Params and GFLOPs of all comparative models in Table XII. Overall, efficiency varies significantly across different architectures. Early CNNs typically utilize fewer parameters, while their GFLOPs vary substantially. For instance, UNet-3D contains merely 6.16-M parameters yet exhibits exceptionally high GFLOPs due to its utilization of 3-D convolutions, which makes it impractical for deployment in resource-constrained environments. Transformer-based methods generally exhibit larger parameter counts and incur higher computational demands, reflecting the costs associated with global attention mechanisms. In contrast, SFSC-CDNet provides a superior efficiency-performance tradeoff. SFSC-CDNet-tiny achieves low computational complexity with 39.90-M parameters and 67.42 GFLOPs, while SFSC-CDNet-small attains higher accuracy with only a moderate increase in GFLOPs. This suggests that the Mamba-based backbone enables capacity expansion under controlled computational growth. The efficiency of SFSC-CDNet stems not only from the Mamba-based backbone, but also from the shared temporal encoding and target-pair interaction strategy, which avoids the computational overhead of constructing multiple independent pairwise networks for multitemporal inputs.

E. Limitations and Future Work

To further clarify the applicability of SFSC-CDNet, this section discusses several aspects that deserve further investigation. SFSC-CDNet shows clear advantages in semantic change discrimination and pseudo-change suppression; however, Recall or Precision is not always the highest on some datasets. This suggests that weakly expressed changes and ambiguous semantic transitions may remain challenging when spectral contrast, texture separability, or boundary evidence is limited, especially in mixed-pixel areas, narrow objects, and irregular transition regions.

In addition, this study mainly validates the proposed framework under bi-temporal and tri-temporal settings. When extended to longer time-series inputs, more complex temporal dependencies, such as gradual, cumulative, or seasonal land-cover transitions, may increase the uncertainty of semantic change inference. Meanwhile, although the framework achieves a favorable accuracy-efficiency tradeoff in terms of GFLOPs, further lightweight optimization is still valuable for edge deployment and large-scale real-time monitoring. Future work will focus on lightweight model design, longer term temporal modeling, and more reliable semantic change inference to further improve robustness, efficiency, and practical applicability.

VI. CONCLUSION

This study introduces SFSC-CDNet, a unified framework for SCD in both bi-temporal and multitemporal remote sensing images. It integrates multidomain difference interaction with bidirectional semantic consistency coupling. By combining spatial semantics with frequency-domain structural cues, the framework mitigates observation perturbations that obscure true changes while amplifying false differences. An edge-guided gated interaction selectively transfers boundary and morphological structures, alleviating contour blurring in cross-temporal fusion. Simultaneously, a bidirectionally coupled decoder with consistency constraints ensures more stable and interpretable semantic correspondence during joint change detection and semantic segmentation. Consequently, the SFSC-CDNet framework demonstrates strong generalization capabilities and achieves a favorable accuracy-efficiency tradeoff, providing consistent benefits across various backbone networks and model scales while maintaining competitive efficiency and deployability.

The main conclusions are as follows.

1) SFSDI enhances change representation by introducing cross-domain complementary cues and a constrained difference metric, effectively reducing biases arising from shifts in imaging conditions and texture perturbations.
2) EGIST improves boundary fidelity by using Gaussian-smoothed discrepancy cues and edge-related transition

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:06:20 UTC from IEEE Xplore. Restrictions apply.

responses to guide gated cross-temporal feature interaction. By facilitating the selective transmission of reliable boundary-related information between bi-temporal features, EGIST enhances boundary localization and contour integrity for fine-grained changes.

3) SCCM addresses objective conflicts by imposing bidirectional constraints between segmentation and change detection during the decoding process, and regulating cross-temporal semantics through consistency regularization, which enhances multitask stability and interpretability.

In conclusion, SFSC-CDNet establishes a collaborative inference paradigm for SCD, supporting more reliable analysis of land-cover transitions in complex multitemporal remote sensing scenarios.

REFERENCES

[1] J. Li, X. Huang, L. Tu, T. Zhang, and L. Wang, "A review of building detection from very high resolution optical remote sensing images," *GISci. Remote Sens.*, vol. 59, no. 1, pp. 1199–1225, Dec. 2022.

[2] Z. Zheng, Y. Zhong, J. Zhao, A. Ma, and L. Zhang, "Unifying remote sensing change detection via deep probabilistic change models: From principles, models to applications," *ISPRS J. Photogramm. Remote Sens.*, vol. 215, pp. 239–255, Sep. 2024.

[3] H. He, J. Yan, D. Liang, Z. Sun, J. Li, and L. Wang, "Time-series land cover change detection using deep learning-based temporal semantic segmentation," *Remote Sens. Environ.*, vol. 305, May 2024, Art. no. 114101.

[4] S. Tian, Y. Zhong, Z. Zheng, A. Ma, X. Tan, and L. Zhang, "Large-scale deep learning based binary and semantic change detection in ultra high resolution remote sensing imagery: From benchmark datasets to urban application," *ISPRS J. Photogramm. Remote Sens.*, vol. 193, pp. 164–186, Nov. 2022.

[5] C. E. Woodcock, T. R. Loveland, M. Herold, and M. E. Bauer, "Transitioning from change detection to monitoring with remote sensing: A paradigm shift," *Remote Sens. Environ.*, vol. 238, Mar. 2020, Art. no. 111558.

[6] W. Jiang, Y. Sun, L. Lei, G. Kuang, and K. Ji, "Change detection of multisource remote sensing images: A review," *Int. J. Digit. Earth*, vol. 17, no. 1, Dec. 2024, Art. no. 2398051.

[7] Y. Li, X. Li, Y. Zhang, D. Peng, and L. Bruzzone, "Cost-efficient information extraction from massive remote sensing data: When weakly supervised deep learning meets remote sensing big data," *Int. J. Appl. Earth Obs. Geoinf.*, vol. 120, Jun. 2023, Art. no. 103345.

[8] Q. Zhu, X. Guo, Z. Li, and D. Li, "A review of multi-class change detection for satellite remote sensing imagery," *Geo-Spat. Inf. Sci.*, vol. 27, no. 1, pp. 1–15, Jan. 2024.

[9] T. Bai et al., "Deep learning for change detection in remote sensing: A review," *Geo-Spat. Inf. Sci.*, vol. 26, no. 3, pp. 262–288, Jul. 2023.

[10] J. Chen, D. Hou, C. He, Y. Liu, Y. Guo, and B. Yang, "Change detection with cross-domain remote sensing images: A systematic review," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 11563–11582, 2024.

[11] Y. Afaq and A. Manocha, "Analysis on change detection techniques for remote sensing applications: A review," *Ecol. Inform.*, vol. 63, Jul. 2021, Art. no. 101310.

[12] Q. Wang, W. Jing, K. Chi, and Y. Yuan, "Cross-difference semantic consistency network for semantic change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4406312.

[13] P. Wang et al., "Enhancing change detection with edge-guided difference modeling in remote sensing imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[14] F. Cui and J. Jiang, "MTSCD-Net: A network based on multi-task learning for semantic change detection of bitemporal remote sensing images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 118, Apr. 2023, Art. no. 103294.

[15] M. Cheng, W. He, Z. Li, G. Yang, and H. Zhang, "Harmony in diversity: Content cleansing change detection framework for very-high-resolution remote-sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 218, pp. 1–19, Dec. 2024.

[16] Z. Zhu, S. Qiu, and S. Ye, "Remote sensing of land change: A multifaceted perspective," *Remote Sens. Environ.*, vol. 282, Dec. 2022, Art. no. 113266.

[17] W. Jing, K. Chi, Q. Li, and Q. Wang, "ChangeRD: A registration-integrated change detection framework for unaligned remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 220, pp. 64–74, Feb. 2025.

[18] R. Zhang, H. Zhang, X. Ning, X. Huang, J. Wang, and W. Cui, "Global-aware Siamese network for change detection on remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 199, pp. 61–72, May 2023.

[19] Y. Liu et al., "Towards resolution-arbitrary remote sensing change detection with spatial-frequency dual domain learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 231, pp. 137–150, Jan. 2026.

[20] R. Caye Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Athens, Greece, Oct. 2018, pp. 4063–4067.

[21] C. Zhang et al., "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020.

[22] L. Ding, J. Zhang, H. Guo, K. Zhang, B. Liu, and L. Bruzzone, "Joint spatio-temporal modeling for semantic change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5610814.

[23] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IGARSS - IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210.

[24] H. Chen, J. Song, C. Han, J. Xia, and N. Yokoya, "ChangeMamba: Remote sensing change detection with spatiotemporal state space model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4409720.

[25] J. Wu and S. Wei, "Remote sensing image change detection method based on difference-guided attention and edge-aware mechanism," *Adv. Space Res.*, vol. 77, no. 5, pp. 5765–5775, Mar. 2026.

[26] J. Lin, G. Wang, D. Peng, and H. Guan, "Edge-guided multi-scale foreground attention network for change detection in high resolution remote sensing images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 133, Sep. 2024, Art. no. 104070.

[27] Z. Zhang, L. Bao, S. Xiang, G. Xie, and R. Gao, "B2CNet: A progressive change boundary-to-center refinement network for multitemporal remote sensing images change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 11322–11338, 2024.

[28] J. Long, S. Liu, M. Li, H. Zhao, and Y. Jin, "BGSNet: A boundary-guided Siamese multitask network for semantic change detection from high-resolution remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 225, pp. 221–237, Jul. 2025.

[29] Y. Tang, S. Feng, C. Zhao, Y. Chen, Z. Lv, and W. Sun, "A semantic change detection network based on boundary detection and task interaction for high-resolution remote sensing images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 9, pp. 17184–17198, Sep. 2025.

[30] Z. Zheng, Y. Zhong, S. Tian, A. Ma, and L. Zhang, "ChangeMask: Deep multi-task encoder-transformer-decoder architecture for semantic change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 183, pp. 228–239, Jan. 2022.

[31] J. Wang, Y. Zhong, and L. Zhang, "Semantic change detection based on supervised contrastive learning for high-resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5649720.

[32] K. Tang, F. Xu, X. Chen, Q. Dong, Y. Yuan, and J. Chen, "The ClearSCD model: Comprehensively leveraging semantics and change relationships for semantic change detection in high spatial resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 211, pp. 299–317, May 2024.

[33] Z. Zhou, K. Hu, Y. Fang, and X. Rui, "SChanger: Change detection from a semantic change and spatial consistency perspective," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 10186–10203, 2025.

[34] Z. Li, X. Wang, S. Fang, J. Zhao, S. Yang, and W. Li, "A decoder-focused multitask network for semantic change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5609115.

[35] S. Fang, W. Li, Y. Song, Z. Li, and J. Zhao, "Rethinking semantic change detection from a semantic alignment perspective," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4417215.

[36] H. Wei et al., "Spatio-temporal feature fusion and guide aggregation network for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:06:20 UTC from IEEE Xplore. Restrictions apply.

[37] J. Ma, J. Duan, X. Tang, X. Zhang, and L. Jiao, "EATDer: Edge-assisted adaptive transformer detector for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5602015.

[38] B. Liu, H. Chen, K. Li, and M. Y. Yang, "Transformer-based multimodal change detection with multitask consistency constraints," *Inf. Fusion*, vol. 108, Aug. 2024, Art. no. 102358.

[39] K. Yang et al., "Asymmetric Siamese networks for semantic change detection in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5609818.

[40] H. Zhang, H. Guo, K. Chen, H. Chen, Z. Zou, and Z. Shi, "FoBa: A foreground–background co-guided method and new benchmark for remote sensing semantic change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5653919.

[41] P. Yuan, Q. Zhao, X. Zhao, X. Wang, X. Long, and Y. Zheng, "A transformer-based Siamese network and an open optical dataset for semantic change detection of remote sensing images," *Int. J. Digit. Earth*, vol. 15, no. 1, pp. 1506–1525, Dec. 2022.

[42] S. Shi et al., "Multi-temporal urban semantic understanding based on GF-2 remote sensing imagery: From tri-temporal datasets to multi-task mapping," *Int. J. Digit. Earth*, vol. 16, no. 1, pp. 3321–3347, Oct. 2023.

[43] L. Ding, H. Guo, S. Liu, L. Mou, J. Zhang, and L. Bruzzone, "Bi-temporal semantic reasoning for the semantic change detection in HR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5620014.

[44] X. Zuo et al., "Multitask Siamese network guided by enhanced change information for semantic change detection in bitemporal remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 61–77, 2025.

[45] Ö. Çiçek, A. Abdulkadir, S. S. Lienkamp, T. Brox, and O. Ronneberger, "3D U-Net: Learning dense, volumetric segmentation from sparse annotation," in *Proc. Int. Conf. Medical Image Comput. Comput.-Assist. Intervent.*, vol. 9901, 2016, pp. 424–432.

[46] R. Caye Daudt, B. Le Saux, A. Boulch, and Y. Gousseau, "Multitask learning for large-scale semantic change detection," *Comput. Vis. Image Understand.*, vol. 187, Oct. 2019, Art. no. 102783.

[47] S. Hou et al., "Semantic-centric change detection framework: Considering spatiotemporal heterogeneity and spatiotemporal correlation of land cover," *Int. J. Digit. Earth*, vol. 18, no. 2, pp. 1–27, 2025.

[48] C. Zhang, L. Wang, S. Cheng, and Y. Li, "SwinSUNet: Pure transformer network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5224713.

[49] Y. Xing, Y. Jia, S. Gao, J. Hu, and R. Huang, "Frequency-enhanced mamba for remote sensing change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[50] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 770–778.

[51] E. Xie et al., "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. Adv. Neural Inf. Process. Sys. (NIPS)*, vol. 34, Dec. 2021, pp. 12077–12090.

[52] X. Liu et al., "GSTM-SCD: Graph-enhanced spatio-temporal state space model for semantic change detection in multi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 230, pp. 73–91, Dec. 2025.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:06:20 UTC from IEEE Xplore. Restrictions apply.